remote sensing

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Review

Review of Recent Advances in Remote Sensing and Machine Learning Methods for Lake Water Quality Management

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Abstract: This review examines the integration of remote sensing technologies and machine learning models for efficient monitoring and management of lake water quality. It critically evaluates the performance of various satellite platforms, including Landsat, Sentinel-2, MODIS, RapidEye, and Hyperion, in assessing key water quality parameters including chlorophyll-a (Chl-a), turbidity, and colored dissolved organic matter (CDOM). This review highlights the specific advantages of each satellite platform, considering factors like spatial and temporal resolution, spectral coverage, and the suitability of these platforms for different lake sizes and characteristics. In addition to remote sensing platforms, this paper explores the application of a wide range of machine learning models, from traditional linear and tree-based methods to more advanced deep learning techniques like convolutional neural networks (CNNs), recurrent neural networks (RNNs), and generative adversarial networks (GANs). These models are analyzed for their ability to handle the complexities inherent in remote sensing data, including high dimensionality, non-linear relationships, and the integration of multispectral and hyperspectral data. This review also discusses the effectiveness of these models in predicting various water quality parameters, offering insights into the most appropriate model–satellite combinations for different monitoring scenarios. Moreover, this paper identifies and discusses the key challenges associated with data quality, model interpretability, and integrating remote sensing imagery with machine learning models. It emphasizes the need for advancements in data fusion techniques, improved model generalizability, and the developing robust frameworks for integrating multi-source data. This review concludes by offering targeted recommendations for future research, highlighting the potential of interdisciplinary collaborations to enhance the application of these technologies in sustainable lake water quality management.

Keywords: remote sensing; machine learning; deep learning; lake water quality

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1. Introduction

Freshwater lakes are indispensable to the ecological health of our planet and play a fundamental role in supporting human life. Covering only about 1.8% of the Earth's surface, freshwater lakes serve as biodiversity hotspots, providing habitats for a wide range of species, including plankton, insects, and fish, all of which contribute to the functioning of aquatic ecosystems [1]. For example, Taihu Lake in China supports diverse macroinvertebrate communities that are crucial for ecosystem productivity and nutrient cycling [2]. Beyond their ecological importance, lakes also bolster regional economies by supporting industries such as fishing, tourism, and water sports. In Ontario, Canada, more than 250,000 lakes contain one-fifth of Earth's freshwater reserves [3], and the region's recreational fishing industry alone generates approximately $7 billion annually while supporting over 75,000 jobs [4]. Lakes further provide significant cultural ecosystem services (CES), including aesthetic, spiritual, and recreational values that enhance human well-being and social connections to nature. For instance, lakes in the European Alps offer

cultural and recreational benefits, contributing to the quality of life in the surrounding regions [5].

Despite their crucial roles, many freshwater lakes, including major ones like Lake Erie in the Great Lakes region of North America, are facing severe degradation in water quality [6]. This degradation is largely due to agricultural runoff, industrial wastewater, and rapid urbanization, which have introduced excessive pollutants into these ecosystems [7–9]. For example, Lake Erie has experienced recurrent harmful algal blooms (HABs), driven by excess nutrients, such as phosphorus and nitrogen, from agricultural activities [10]. These blooms significantly impact water quality, endanger drinking water supplies, and threaten aquatic life, leading to substantial economic losses for industries dependent on the lake, including fishing and tourism. For example, harmful algal blooms in Lake Erie alone have been estimated to cost the regional economy $272 million annually, affecting both the fishing and tourism sectors [11].

Amid growing pressures from population growth, industrial activities, and climate change, accurately predicting and monitoring lake water quality has become critical. Such efforts are essential to ensure access to safe drinking water, preserve biodiversity, and support sustainable development goals [12–14]. Robust monitoring systems provide valuable data for managing freshwater resources and developing mitigation strategies to combat pollution and habitat degradation.

Monitoring lake water quality presents unique challenges compared to rivers and coastal systems. Lakes typically have more static conditions than rivers, with limited water flow, leading to complex stratification and mixing processes that create significant spatial heterogeneity in water quality [15,16]. Water quality parameters can exhibit significant spatial heterogeneity due to various environmental factors. For instance, coastal areas [17,18] are influenced by tidal movements and interactions between freshwater and saltwater, creating variability in water quality. By contrast, lakes often require higher resolution data to capture their complex internal dynamics and seasonal changes, making them more challenging to monitor. Addressing these complexities through advanced monitoring techniques is vital for both environmental sustainability and public health protection.

2. Historical Development of Remote Sensing for Lake Water Quality

While in situ monitoring provides accurate and direct measurements of water quality parameters, it faces several challenges, including high costs [19], limited spatial coverage [20], and logistical difficulties in accessing remote or hazardous areas [21,22]. Furthermore, the temporal frequency of data collection is often insufficient to capture rapid changes in water quality, limiting timely responses to environmental threats [23]. To address these limitations, remote sensing technologies, especially satellite-based sensors, have opened new frontiers in environmental monitoring, including not only water quality, but also agricultural management, deforestation tracking, and urban heat island studies. For example, NDVI, which developed the use of red and near-infrared bands, has been widely applied to monitor global vegetation health and agricultural productivity [24]. In deforestation studies, MODIS data have been crucial for mapping forest cover changes in the Amazon, identifying key areas of illegal logging [25]. Additionally, Sentinel-2 data have been used extensively in urban heat island studies, providing valuable thermal information to guide heat mitigation strategies in cities like Los Angeles [26]. Given these advancements, remote sensing technologies have also played a transformative role in water quality prediction, enabling large-scale and continuous monitoring of key parameters such as chlorophyll-a, turbidity, and suspended sediments [27–31].

The use of remote sensing for water quality assessment began as early as the 1970s, with platforms like Landsat enabling the monitoring of large water bodies [32,33]. Initial studies focused on retrieving basic water quality parameters, such as chlorophyll-a, turbidity, suspended solids [34], and temperature [28], using multispectral data. However, these early applications were constrained by several factors, such as the relatively low spectral and spatial resolution of satellite sensors and the reliance on empirical models that required

ground-based calibration. For example, empirical models were commonly used to correlate spectral data with water quality indicators, but these models often lacked generalizability due to their sensitivity to local environmental conditions and sensor limitations [35].

Another limitation of early remote sensing methods was their inability to account for the complex interactions between different water quality components, such as the influence of suspended sediments on chlorophyll reflectance signals. These limitations often led to inaccuracies in water quality predictions, especially in optically complex waters where multiple constituents could interfere with the spectral signals [36,37].

With the advent of new satellite technologies in the 1990s and 2000s, such as MODIS and MERIS [38–40], the field of water quality monitoring experienced significant progress. These platforms offered improved spatial and spectral resolution, allowing researchers to detect more subtle variations in water quality over time and space [41–43]. For instance, the enhanced spectral resolution of MERIS, which was specifically designed for water applications, enabled more accurate monitoring of chlorophyll-a concentrations and detection of harmful algal blooms (HABs) in European lakes [44].

One key advancement during this period was the development of semi-analytical models that allowed for the retrieval of multiple water quality parameters simultaneously [45,46]. These models provided more flexibility compared to earlier empirical approaches, as they could be applied across different geographic regions and sensor types.

The 21st century saw a paradigm shift in water quality monitoring with the integration of machine learning algorithms into remote sensing applications [47]. Machine learning models, such as random forests, support vector machines, and deep learning models like convolutional neural networks (CNNs), have been successfully employed to process vast amounts of remote sensing data and extract meaningful patterns from high-dimensional datasets [48–50].

These models are particularly effective in handling non-linear relationships between spectral data and water quality parameters, which traditional linear models often fail to capture. For example, Cao et al. [51] applied a modified particle swarm optimization algorithm coupled with partial least squares (PLS) regression to retrieve chlorophyll-a and turbidity from hyperspectral imagery with high accuracy in optically complex waters. Similarly, neural networks have been employed to predict water quality parameters in real-time by continuously learning from large datasets, enabling more timely and efficient decision-making in water management [52].

High spectral resolution technologies, like hyperspectral imaging, have been applied to water quality monitoring since the early 2000s [53], with advancements in sensors and data processing enhancing their capabilities over time. Recent technologies, such as PRISMA and DESIS, continue to push the boundaries of hyperspectral imaging by offering improved spectral resolution, allowing for more precise monitoring of multiple water quality indicators, including nutrient concentrations and phytoplankton biomass [54]. The integration of hyperspectral data with machine learning models further refines predictions by extracting detailed information from complex water systems, improving accuracy across various environmental conditions.

Remote sensing and machine learning offer immense potential for water quality prediction. By combining satellite data with machine learning, researchers can develop real-time monitoring systems that inform policy, guide resource management, and support emergency responses [55,56]. As these technologies evolve, they will enable more accurate and timely assessments. Future research will focus on multi-sensor data fusion to overcome individual sensor limitations and enhance temporal resolution, while cloud-based real-time analytics will improve early detection of harmful algal blooms and other water quality threats [57]. However, there are challenges, such as data heterogeneity, varying resolutions, and high computational demands, which require multidisciplinary collaboration between environmental science, AI, and data engineering to overcome [58,59].

The importance of water quality prediction is underscored and driven by the necessity to safeguard public health, preserve ecosystems, and manage water resources

sustainably [60,61]. Advances in remote sensing and machine learning technologies have transformed the field, providing new tools and methods for comprehensive monitoring and prediction. While challenges remain, integrating these technologies holds great promise for enhancing our ability to predict and manage water quality effectively. This comprehensive review aims to consolidate and synthesize recent advancements in machine learning and remote sensing for predicting lake water quality. By integrating findings from various studies, this review highlights the most effective methodologies, identifies common challenges, and suggests avenues for future research. The goal is to critically evaluate how recent advances in technology can improve efficiency and reduce the cost of lake water quality monitoring programs around the globe, offering insights that could support environmental management and policymaking.

3. Advanced Remote Sensing Techniques for Water Quality Prediction

This section delves into the various remote sensing techniques employed for predicting water quality of lakes. It begins with a discussion of preprocessing methods that ensure the precision and dependability of remote sensing data by addressing specific issues, including atmospheric correction, geometric correction, radiometric correction, noise reduction, and cloud masking. Subsequently, it explores the prediction of key water quality variables, emphasizing the distinction between optical and non-optical active variables and the effectiveness of remote sensing techniques in monitoring these parameters.

3.1. Lake Water Quality Parameters

Using remote sensing data, key water quality parameters, including optical and non-optical active variables (TP [62–65], TN [52,66–68] and pH [69–71]), can be predicted. Recent research developments utilizing remote sensing techniques for lake water quality monitoring, as illustrated in Table 1, highlight substantial progress in predicting a range of parameters, including chlorophyll-a (Chl-a), turbidity, colored dissolved organic matter (CDOM), water temperature, as well as lake water levels and surface area. These studies utilize data from satellite sources like Sentinel-2, Landsat-8, and MODIS, applying advanced predictive models, including random forests (RFs), gradient boosting decision trees (GBDTs), neural networks, and stepwise regression. The spectral bands used range widely, with specific bands tailored to each parameter, ensuring accurate predictions. Evaluation metrics such as R^2, *RMSE*, and accuracy rates highlight the effectiveness of these models, achieving high correlations and low error rates, thereby supporting the capacity of remote sensing in providing reliable, large-scale water quality assessments. Optical active variables, such as Chl-a, turbidity, CDOM, and temperature play a crucial role in assessing lake water quality. These variables can be directly detected using remote sensing because of their distinctive spectral properties. Recent advancements in satellite technology and machine learning algorithms have meaningfully improved the accuracy and resolution of these measurements, enabling more effective monitoring of lake water ecosystems.

Chlorophyll-a serves as a key indicator of phytoplankton biomass and a proxy for eutrophication and algal blooms. The most commonly used spectral bands for Chl-a estimation include blue, green, red, and NIR [72–76]. Some studies have incorporated UV bands [77], while others have focused solely on red and red-edge bands [78]. The inclusion of UV bands helps in capturing specific absorption features of Chl-a, while the red and red-edge bands are particularly effective due to their strong absorption and reflectance properties. Various remote sensing sensors have been utilized to monitor Chl-a concentrations [72–88], including the widely studied American satellites Landsat 5, 8 [72,76], and MODIS [73], and European satellites Sentinel-2 [74,79–81,84,86] and RapidEye [79]. Different predictive models have been used to predict Chl-a concentrations. These include band ratio empirical models [72,79], deep chlorophyll-a maxima (DCMs) [73], random forest (RF) [74,76], and cubist [75]. Band ratio empirical models use the ratio of different spectral bands to assess Chl-a concentration. They are simple and effective, especially when the relationship between spectral reflectance and Chl-a is linear. These models,

combined with high-resolution satellite data, enabled detailed and accurate mapping of Chl-a concentrations across large spatial scales.

Turbidity measures water clarity and is affected by the concentration of suspended particles in the water column. It affects light penetration and is critical for aquatic ecosystems. Remote turbidity sensing typically uses the visible (400–700 nm) and NIR bands, as well as SWIR 2 and red-edge bands. The visible and NIR bands are effective because they capture the scattering of light by suspended particles, which is directly related to turbidity. SWIR 2 and red-edge bands are used because they can penetrate the water column more deeply and are less affected by surface reflection, providing more accurate measurements of turbidity in various water depths [79,87–93]. Predictive models, including the super learner algorithm (SLA) and multilayer perceptron (MLP) [89], gradient boosting decision trees (GBDT) [87], band ratio empirical models [79], normalized difference turbidity index (NDTI) empirical models [92], and ensemble deep learning approach (e-DLA) [93], have been applied to estimate turbidity concentrations. These models are chosen for their capacity to manage intricate, non-linear relationships (MLP and SLA) [89], capture intricate patterns (GBDT) [87], provide quick estimations (band ratio) [79], enhance contrast between turbid and clear water (NDTI) [92], and achieve high accuracy rates in detailed and large-scale monitoring (e-DLA) [93], thereby significantly improving the accuracy and efficiency of turbidity monitoring through advanced remote sensing technology.

CDOM affects watercolor and plays a role in nutrient cycling and water chemistry. It primarily absorbs ultraviolet (UV) and blue light, with significant absorption features around 254 nm and 365 nm. Remote sensing of DOM utilizes these absorption characteristics, often through hyperspectral imagery [94,95]. Algorithms for CDOM estimation include empirical relationships between absorption coefficients and CDOM concentrations and more advanced models using hyperspectral data to capture specific absorption features.

Water temperature is a crucial parameter influencing chemical reactions, biological processes, and physical properties of lakes. Remote water temperature sensing uses thermal infrared (TIR) bands, typically in the 10.4–12.5 μm range. Sensors such as the Thermal Infrared Sensor (TIRS) on Landsat 8 and MODIS deliver surface temperature data featuring fine spatial and temporal resolution. Studies [96,97] have shown that thermal infrared remote sensing can effectively monitor water temperature dynamics, providing valuable data for understanding lakes' thermal pollution and stratification processes. Hulley and Hook [96] presented temperature emissivity separation (TES) and split-window algorithms to ensure that emissivity data from ASTER and MODIS sensors can be used interchangeably, providing consistent and reliable data. A recent study [97] assessed the effectiveness of the single-channel, split-window, and multi-channel algorithms in assessing lake surface water temperature (LSWT) using data from Landsat 8, showing that the split-window algorithm generally performed the best for LSWT estimation due to its effective correction for atmospheric effects.

Lake surface area detection using remote sensing is essential for accurately defining the spatial extent of water bodies, which is fundamental for subsequent water quality parameter estimations. This process involves identifying a lake's boundaries and surface area, ensuring that pixel-based water quality data is accurately allocated and that land features are not misclassified as water. The significance of water body detection lies in its ability to enhance the precision of water quality predictions by ensuring only water pixels are analyzed, thus improving the reliability of derived parameters. Recent research has employed various detection methods, including thresholding techniques on spectral bands, water indices like the normalized difference water index (NDWI), and advanced machine learning algorithms. For instance, the NDWI utilizes green and near-infrared bands to improve the contrast of water features against vegetation and soil [98]. Machine learning models, such as RF, support vector machines (SVM), and change vector analysis (CVA), have also been applied to improve detection accuracy by incorporating the spectral, spatial, and textural features of a body of water [99,100]. Additionally, deep learning methods, such as convolutional neural networks (CNNs), have been utilized to enhance water body

detection through semantic segmentation techniques [98,101,102]. These advancements underscore the importance of precise water body detection in remote sensing-based water quality monitoring, as they significantly contribute to the precision and dependability of water quality assessments.

Lake water level monitoring is crucial for understanding hydrological changes, assessing flood risks, and managing water resources. Remote sensing techniques offer the ability to provide regular and spatially comprehensive measurements for water level monitoring. Typically, water level predictions use LiDAR remote sensing to measure baseline water depths and determine the corresponding lake surface extent. Subsequently, optical satellite data from different time points are used to analyze a lake's surface area at the specific target time, allowing for water level calculations by combining lake area changes with digital elevation models (DEMs) [103,104]. This approach simplifies the estimation of water levels across various spatial locations within a lake by only requiring surface area measurements. However, it is heavily dependent on the accuracy of the elevation models. Inaccurate DEMs can lead to significant errors in water level predictions.

In addition to the indirect method that relies on DEMs, direct water level monitoring using optical satellite imagery has also gained attention [105,106]. This method calculates water levels directly from spectral images without depending on elevation models, offering a simpler and more direct approach. The accuracy of this method is strongly related to the bathymetric capabilities of the satellites when applied to deeper or turbid lakes, where light penetration is limited. For example, fusing data from Landsat and ICESat-2 data were employed to monitor the water levels of Lake Mead, successfully determining annual variations [106]. The study demonstrated that ICESat-2 has significant capability to precisely map Earth's surface topography, successfully capturing signal photons bouncing off underwater surfaces down to a depth of about 10 m in Lake Mead. Furthermore, in research [105] of East Dongting Lake, where water depths exceed 20 m and the distribution of water and sediment is irregular across the area, researchers faced significant challenges due to the influence of suspended matter. In this case, GF-1 multispectral reflectance was employed to simulate water depth, and green and red bands combination were chosen to build a dual bands model, which achieved a correlation coefficient as high as 0.925.

Traditional water level inversion models often struggle with non-linear relationships and mixed signals from complex surface conditions (e.g., vegetation or high turbidity), limiting the accuracy of water level estimations. To address these limitations, recent advancements have integrated machine learning models with remote sensing data. Machine learning algorithms, such as RF, SVM, and deep learning neural networks, have been employed to model complex interactions between remote sensing imagery and hydrological parameters. These methods have demonstrated superior performance in capturing non-linear dependencies and adjusting for regional variability, as evidenced by studies on water bodies such as Hulun Lake [107], Poyang Lake [108], and the Tibetan Plateau [109].

Table 1 provides a comparison of predictive spectral data for water quality parameters. In Table 1, coefficient of determination (R^2), root mean square error (*RMSE*), and mean absolute error (*MAE*) are selected as the primary metrics to assess model performance. For each reference, when multiple R^2, *RMSE*, or *MAE* values are reported for different models, the best values are selected.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(x_i - y_i)^2}{\sum_{i=1}^{n}(\overline{y} - y_i)^2} \quad (1)$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i-1}^{n}(x_i - y_i)^2} \quad (2)$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|x_i - y_i| \quad (3)$$

where y_i, x_i, and $\overline{y}$ represent the actual, the predicted, and the average value for the i-th target variable. The term n represents the total number of samples.

Table 1. Comparison of predictive spectral data for water quality parameters.

Parameter	Data Source							Spectral Bands												Ref.	Evaluation Metrics		
	a	b	c	d	e	f	g	h	A	B	C	D	E	F	G	H	I	J	K	L		R^2	*RMSE*	*MAE*
Chl-a	√		√								√	√	√		√						[72]	0.62		
			√		√						√	√	√	√	√	√	√	√			[89]	0.71	6160 µg/L	4970 µg/L
				√							√	√	√		√						[73]	0.71	0.8 µg/L	
					√						√	√	√		√						[74]	0.90	70 µg/L	
				√								√	√								[75]	0.40	0.34	
	√		√								√	√	√		√						[76]	0.56		15.2 µg/L
						√						√		√							[79]	0.37		
	√	√	√								√	√	√		√	√	√				[110]	0.99	0.34 µg/L	0.07 µg/L
			√		√			√		√	√	√	√	√	√	√	√	√			[70]	0.92		
	3	1	5	1	3	1	0	1	0	1	7	9	8	3	7	3	3	2	0	0				
Turbidity			√		√						√	√	√	√	√	√	√	√			[89]	0.82	2.05 NTU	1.10 NTU
					√							√	√	√			√				[87]	0.88	9.90 NTU	6.71 NTU
						√						√	√								[79]	0.92		
	√		√								√	√	√		√						[92]	0.99		0.31 NTU
			√							√		√			√	√	√				[69]	0.99		
	1	0	3	0	2	1	0	0	0	1	2	5	4	2	3	2	3	1	0	0				
CDOM					√						√	√	√								[94]	0.81		
				√							√	√	√								[95]	0.55	0.04 mg/L	
	0	0	0	1	1	0	0	0	0	0	2	2	2	0	0	0	0	0	0	0				
Water Temp.			√																√	√	[97]	0.95	1.66 °C	
				√															√	√	[66]	0.90	0.10 °C	
	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	2				
Lake Water Level								√				√	√								[105]	0.93		
	√	√						√				√				√	√				[106]	1.00	1.06 m	
				√				√			√		√		√						[103]			
	1	0	0	1	0	0	0	3	0	0	3	2	3	0	2	1	1	0	1	1				
Lake Surface Area					√						√	√	√		√		√				[98]		mIoU = 74%	
								√			√	√	√		√						[99]			
				√				√			√	√	√		√						[100]	0.88	38.45 km²	
	0	0	0	1	1	0	0	2	0	0	3	3	3	0	3	0	1	0	0	0				

The data sources are represented by the following letters: a for Landsat 5, b for Landsat 7, c for Landsat 8, d for MODIS (Aqua/Terra), e for Sentinel-2A/B, f for RapidEye, g for Hyperion, and h for others (Göktürk-2 [70], GF-1 [105], EO1-ALI [106], AVHRR [103], Jason-1/-2/-3 [100], and GRACE [100]). The spectral bands are denoted by the letters A to L: A for ultraviolet, B for ultra blue, C for blue, D for green, E for red, F for red-edge, G for NIR (near-infrared), H for SWIR 1 (short-wave infrared 1), I for SWIR 2 (short-wave infrared 2), J for narrow NIR, K for TIR 1 (thermal infrared 1), and L for TIR 2 (thermal infrared 2). And mIoU evaluates the overlap between predicted lake surface area and true area, which is crucial for segmentation tasks in remote sensing. "√" indicates the presence or use of the corresponding data source, spectral band, or evaluation metric for the listed water quality parameter.

Chlorophyll-a detection shows strong predictive capabilities when using Sentinel-2 [70,74,89], RapidEye [110], and Landsat [70,72,76,89,110]. Studies report high R^2 values, particularly when advanced models such as RFs and gradient boosting trees (GBTs) are employed [70,110]. The inclusion of near-infrared (NIR) and red-edge bands in addition to the visible spectrum enhances the effectiveness of these platforms in capturing the spectral characteristics of chlorophyll-a, as NIR bands are sensitive to vegetation and algal presence in water [70,72–79,110,111]. This combination of bands allows for more accurate estimation of chlorophyll-a concentrations across varying environmental conditions.

Turbidity monitoring relies heavily on green, red, red-edge, NIR, and SWIR bands, which are crucial for detecting suspended particulate matter in water. Landsat 8 [69,89,92,93] and Sentinel-2 [89] have been particularly effective in turbidity prediction due to their spectral resolution and band availability. Advanced models, such as GBT [87] and multilayer perceptron (MLP) [89], further enhance prediction accuracy by capturing the intricate non-linear relationships inherent in turbidity data. The combination of these satellites and models allows for the precise monitoring of turbidity, particularly in waters with high sediment concentrations.

CDOM retrieval typically leverages the blue, green, and red bands, which are sensitive to the absorption characteristics of dissolved organic materials. Sentinel-2 [94] provides superior performance for CDOM detection due to its higher spatial resolution, allowing for more detailed observations. With its high temporal resolution, MODIS [95] is also effective, particularly in large-scale monitoring where frequent data collection is essential. The choice of satellite often depends on the scale and resolution required for the study, with Sentinel-2 excelling in more localized, detailed assessments, while MODIS is preferred for broader, more frequent observations.

Water Temperature monitoring relies on the thermal infrared (TIR) bands of Landsat and MODIS, which measure the emitted radiation from the lake surface. These bands are crucial for accurate temperature estimation, with studies reporting high R^2 values of 0.95 and 0.90 [66,97], and *RMSE* value of 1.66 °C and 0.10 °C. Landsat's TIR bands provide the necessary spatial resolution for detailed temperature assessments, while MODIS offers higher temporal resolution, making it suitable for continuous monitoring. The combination of these satellite data ensures reliable temperature measurements across different water bodies, contributing to effective environmental monitoring.

Lake water level and surface area detection primarily relies on visible and NIR bands [98–100,103,104]. Recent advancements have introduced methods for predicting water levels directly from spectral imagery [105,106], although their accuracy can be affected by factors such as water depth and turbidity. The integration of remote sensing data with advanced modeling techniques provides a comprehensive framework for monitoring lake water quality, water body area, and water levels. By choosing the right combination of satellite data and predictive models, researchers can enhance the accuracy of monitoring and forecasting these parameters, which in turn aids more well-informed decision-making in managing water resources and protecting environment.

The choice between remote sensing reflectance (Rrs), top-of-atmosphere reflectance (Rrc), and intrinsic optical properties (IOPs) depends largely on the water body's optical properties and the atmospheric conditions. Rrs is widely used for retrieving chlorophyll-a concentrations, utilizing bands in the blue, green, and red spectra due to Chl-a's distinctive absorption features [72,73]. Studies [75] have shown that Rrs is particularly effective in clearer water systems where atmospheric interference is minimal. However, its accuracy can decrease in optically complex or turbid waters due to greater atmospheric and particulate interference.

Rrc is more commonly employed in more optically complex waters, such as those with high turbidity or suspended solids. Rrc accounts for atmospheric interference, especially in regions where high aerosol concentrations can distort surface signals. Studies have applied Rrc in turbid waters to improve the reliability of water quality retrievals by mitigating atmospheric distortion [92,112].

IOPs can accurately reflect the absorption and scattering characteristics of water, making them suitable for retrieving more complex parameters in turbid waters, such as backscattering coefficients, absorption coefficients, suspended particulate matter, and colored dissolved organic matter (CDOM) [113,114].

3.2. Remote Sensing Platforms

Different remote sensing platforms provide unique advantages for lake water quality prediction. These platforms include UAVs, aircraft, and satellites, each offering varying levels of spatial and temporal resolution levels.

UAVs provide high-resolution images and can be deployed for targeted monitoring of small to medium-sized lakes [115]. UAVs are particularly useful for capturing detailed spatial variability in water quality parameters. Studies [116–118] have used UAVs equipped with multispectral sensors to monitor water quality parameters like Chl-a and turbidity in lakes, demonstrating high spatial resolution and flexibility. UAVs offer the flexibility to conduct frequent monitoring missions over targeted areas, making them ideal for assessing water quality in smaller or hard-to-reach water bodies [70,119].

Aircraft cover larger areas than UAVs and can carry advanced sensors for detailed spectral analysis. Regional water quality studies often use aircraft to gather high-resolution data over larger spatial extents. For instance, aircraft with hyperspectral sensors have been employed to map dissolved organic matter and other water quality parameters in large lakes [120–122]. However, aircraft operations are more costly and require more workforce compared to UAVs, making UAVs a more cost-effective option for many applications.

Satellites offer extensive spatial and temporal coverage, making them ideal for large-scale monitoring. Commonly used satellites include the Landsat series [123,124], Sentinel [125,126], and MODIS [127,128]. These satellites provide regular, repeatable coverage of large areas, which is crucial for long-term monitoring of water quality trends. Satellite data have been extensively used to monitor parameters like Chl-a and turbidity in lakes worldwide. Table 2 summarizes the parameters of different satellites and their applications in water quality monitoring.

Table 2. Spectral Band Applications in Water Quality Parameters Assessment.

Satellite	A	B	C	D	E	F	G	H	I	J	K	L	Parameters	Refs.
Landsat 5			√	√	√		√	√	√		√		Chl-a, Turbidity	[71,92,93]
Landsat 7			√	√	√		√	√	√		√		CDOM, Chl-a	[129,130]
Landsat 8		√	√	√	√		√	√	√		√		CDOM, Chl-a, Turbidity	[131,132]
MODIS	√		√	√	√		√	√	√		√	√	Chl-a	[133–136]
Sentinel-2		√	√	√	√	√	√	√	√	√			CDOM, Chl-a, Turbidity	[126,137–144]
RapidEye			√	√	√	√	√						CDOM, Chl-a, Turbidity	[145–147]
Hyperion	√	√	√	√	√	√	√	√	√	√			Chl-a	[143,144]

Spectral bands are denoted by the letters A to L: A for ultraviolet, B for ultra blue, C for blue, D for green, E for red, F for red-edge, G for NIR (near-infrared), H for SWIR 1 (short-wave infrared 1), I for SWIR 2 (short-wave infrared 2), J for narrow NIR, K for TIR 1 (thermal infrared 1), and L for TIR 2 (thermal infrared 2). "√" indicates the presence of the corresponding spectral band for the listed satellite.

When evaluating the performance and suitability of these satellites for lake water quality monitoring, several key factors must be considered, including spatial resolution, temporal resolution, spectral bands, and technological innovations in the onboard sensors. The operational timeline depicted in Figure 1 underscores the progression from earlier satellite missions such as Landsat 5 and Landsat 7, which provided foundational datasets essential for historical trend analysis, to more advanced systems like Sentinel-2 and MODIS. These latter platforms offer enhanced spectral and spatial resolutions, as well as more frequent revisit times, thus catering to the demands of modern, high-resolution environmental monitoring. Table 2 further elaborates on the spectral band presence across these satellites, directly linking to their respective water quality monitoring capabilities. Landsat 5 and 7, operational during the late 20th and early 21st centuries, utilized basic visible and near-infrared bands to monitor parameters such as Chl-a and turbidity. These missions laid the groundwork for understanding large-scale environmental processes and provided baseline data that remain invaluable for retrospective analyses.

Landsat 8, launched in 2013, significantly enhanced the capabilities established by its predecessors through the addition of ultra blue and thermal infrared bands. These bands have enabled more detailed assessments of water quality, including the monitoring of TP [148] and refined turbidity metrics [131], making Landsat 8 a critical tool for ongoing aquatic ecosystem studies. Landsat 8 also includes two thermal bands with a 100 m spatial resolution, enhancing its capability to monitor surface water temperature [97], a critical parameter for many ecological processes. This makes Landsat 8 well-suited for large lakes where historical data is essential for understanding long-term trends and for lakes where temperature monitoring is a priority. The revisit cycle of 16 days for these satellites allows for regular monitoring, though this may be less frequent than for some other satellites. However, the long history of Landsat data is invaluable for long-term trend analysis, making them particularly useful for large lakes where long-term environmental changes are being tracked [123].

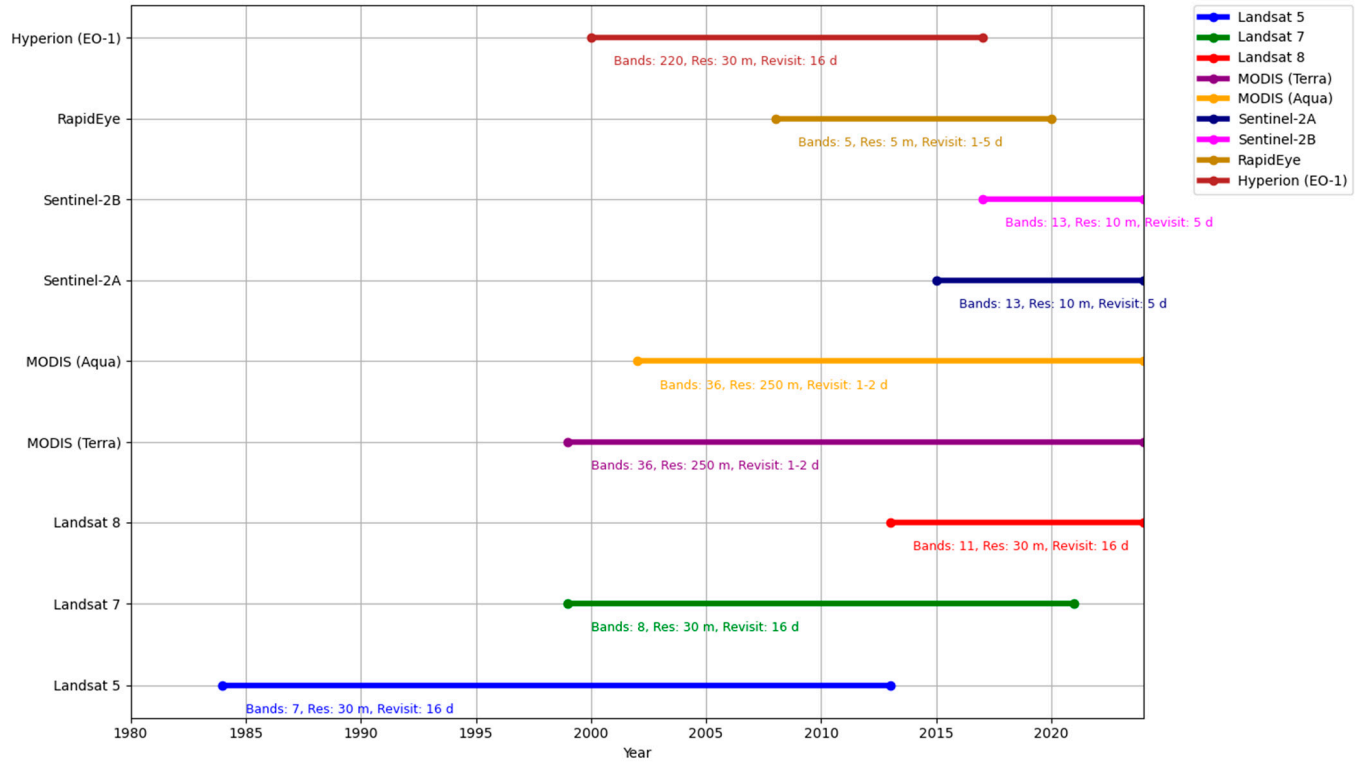


Figure 1. Comparison of satellite specifications.

MODIS, with its sensors aboard the Terra and Aqua satellites, offers lower spatial resolutions of 250 m, 500 m, and 1 km. Despite its lower spatial resolution compared to other satellites, MODIS compensates with a high temporal resolution, providing daily revisits, and a broad range of spectral bands, including multiple thermal infrared bands. These abilities are beneficial for capturing short-term changes and large-scale events, such as water surface temperature changes [133–136], algal blooms [43,149–152] and sediment plumes [43,150,151], making MODIS highly suitable for very large lakes or reservoirs where frequent observations are crucial. This capability is indispensable for observing large water bodies where changes can occur rapidly.

Sentinel-2A and Sentinel-2B, operational since 2015 and 2017, respectively, represent a significant advancement in satellite-based water quality monitoring. These platforms offer high spatial resolution (10 m) and a comprehensive set of spectral bands, including red-edge and short-wave infrared (SWIR) bands. The inclusion of these bands is crucial for detecting subtle changes in water quality, such as variations in CDOM [126,138–141] and TSS [139,153]. The high revisit frequency (5 days) of Sentinel-2 further enhances its utility in monitoring more sensitive water bodies, where timely detection of quality changes is essential [126,137–142]. The advanced multispectral instrument (MSI) on Sentinel-2 enhances its capability to provide detailed and accurate data, making it highly effective for small to medium-sized lakes [80,126] with significant spatial and temporal variability.

RapidEye, although no longer operational, provided data with high spatial resolution (5 m) and limited spectral bands, including red-edge. This made it particularly suitable for detailed studies of specific water quality parameters in smaller water bodies [79,145–149]. RapidEye's data continues to be valuable for studies requiring fine spatial resolution.

Hyperion, aboard the EO-1 satellite, through its hyperspectral imaging capabilities, offered an unparalleled level of spectral detail across 220 bands ranging from 400 to 2500 nm, with a 30-m spatial resolution. This extensive spectral range allowed for highly specific water quality assessments of medium to large lakes [143,144,154]. Despite its decommissioning in 2017, Hyperion's data remains crucial for retrospective analyses and

serves as a benchmark for developing models applied to current hyperspectral data from other platforms.

The effectiveness and applicability of various platforms for lake water quality monitoring are inherently tied to the specific characteristics of each lake and the water quality parameters of interest. Satellites with different spatial and temporal resolutions, spectral bands, and operational timelines each play vital roles in this domain.

UAVs provide high-resolution, flexible monitoring options for small to medium-sized lakes, offering detailed insights into spatial variability in parameters [115–118]. Their ability to conduct frequent, targeted missions makes them ideal for hard-to-reach or smaller water bodies [70]. Aircraft, on the other hand, cover larger areas and are equipped with advanced sensors for detailed spectral analysis, making them suitable for regional water quality studies over large lakes [120–122]. However, the higher cost and operational demands of aircraft make UAVs a more cost-effective alternative for many applications.

Landsat satellites (e.g., Landsat 5, 7, and 8), with their long operational history and moderate spatial resolution, are ideal for large lakes where long-term trend analysis is critical. These satellites are particularly effective for monitoring parameters such as Chl-a [155] and surface temperature [97]. Offering higher spatial resolution and more frequent revisit times, Sentinel-2 is better suited for medium-sized lakes and dynamic water quality parameters like CDOM [137,138,141] and TSS [139]. By contrast, MODIS, with its broad spectral coverage and frequent revisits, excels in monitoring very large lakes with parameters requiring frequent observation, such as surface temperature [133–136] and algal blooms [43,149–152]. With its fine spatial resolution, RapidEye was particularly suitable for small lakes or localized studies [79,145,149].

Multispectral data from satellites, like Landsat [156] and Sentinel [116], are widely utilized for general monitoring, offering accessibility and broad spectral coverage for parameters like turbidity and chlorophyll concentration [157]. However, its broader bands may lack the specificity needed for detecting subtle changes. By contrast, hyperspectral data from sensors like Hyperion [158] and PRISMA [159] provide detailed spectral information, enabling identification of specific pollutants [160,161]. Despite its higher complexity and cost, hyperspectral data are essential for studies requiring high precision.

Spatial resolution is crucial for accurate water quality monitoring, especially in nearshore areas where pollution concentrations can be highly localized. Sentinel-2's high spatial resolution (10 m) allows for detailed observations, which are essential for managing pollution sources effectively [126,137–142]. Conversely, satellites with lower spatial resolutions, such as MODIS, may not capture such localized pollution events, making them less suitable for nearshore monitoring where fine-scale spatial detail is needed.

Temporal resolution is equally important, particularly for parameters that change rapidly, such as dissolved oxygen, which can fluctuate significantly within hours due to factors like temperature and biological activity. Satellites with higher temporal resolution, like MODIS [43,149–152,162], which offers daily revisits, are better suited for monitoring these dynamic parameters. By contrast, satellites with longer revisit cycles, such as Landsat (16 days), may struggle to capture rapid changes effectively. Aligning temporal resolution with the frequency of water quality changes is critical for accurate and timely monitoring.

For satellites that have been decommissioned, such as Landsat 5 and Landsat 7, their historical data remains invaluable for ongoing research. During their operational periods, these satellites provided real-time water quality assessments crucial for monitoring environmental changes. Post-decommissioning, their datasets continue to serve as essential resources, often integrated with data from currently operational satellites to enhance analysis and extend the temporal scope of studies [71,72,76,92,93,110].

3.3. Spectral Indices for Water Quality Assessment

The effectiveness of spectral indices in lake water quality assessment is primarily rooted in their ability to highlight specific spectral features associated with different water quality parameters. Table 3 summarizes several key spectral indices used in lake water

quality assessment, including their calculation formulas, satellite bands used, and the specific water quality parameters they predict. This section delves deeper into the principles behind each index and discusses why they are successful in predicting various water quality parameters, supported by relevant examples.

Table 3. Spectral Indices and Applications in Lake Water Quality Assessment.

Index	General Formula	Dataset	Parameters	Ref.
NDVI	$\frac{NIR-Red}{NIR+Red}$	Landsat TM Landsat 8 Sentinel-2	Chl-a Temperature Turbidity	[163] [164] [165]
NDTI	$\frac{Red-Green}{Red+Green}$	Sentinel-2	Turbidity	[166]
EVI	$\frac{2.5\times(NIR-Red)}{NIR+6\times Red-7.5\times Blue+1}$	MODIS Landsat 8	Chl-a Water Level	[167] [167]
NDWI	$\frac{Green-NIR}{Green+NIR}$	Sentinel-2	Turbidity Chl-a Water body	[168] [167] [169]
NDCI	$\frac{RedEdge1-Red}{RedEdge1+Red}$	Sentinel-2	Chl-a	[170]
MCI	$\frac{RedEdge2-RedEdge1}{RedEdge2+RedEdge1}$	Sentinel-2	Chl-a	[167]
SCI	$\frac{NIR-RED}{NIR+Red}$	Landsat TM	Chl-a	[171]

NDVI (Normalized Difference Vegetation Index): NDVI takes advantage of chlorophyll's effective absorption of red light and the high reflectance of near-infrared (NIR) light by healthy vegetation. This contrast allows NDVI to effectively monitor vegetation health and density, making it highly suitable for gauging Chl-a concentrations in lakes [163,164]. The red (Band 4) and NIR (Band 8) bands of Sentinel-2 are particularly sensitive to chlorophyll, which is why NDVI is widely used for detecting algal blooms and assessing the trophic state of lakes. High NDVI values indicate dense vegetation, which correlates with high Chl-a levels, making it a robust indicator of algal biomass.

NDTI (Normalized Difference Turbidity Index): NDTI focuses on the difference in reflectance between red and green light to estimate turbidity, a measure of water clarity affected by suspended particles. Turbidity causes scattering and absorption, particularly in the red and green wavelengths. Sentinel-2's red (Band 4) and green (Band 3) bands are ideal for this purpose, as they capture the optical properties affected by suspended materials. By contrasting these bands, NDTI emphasizes areas with higher concentrations of suspended solid suspended solids concentrations. This makes NDTI particularly effective for real-time monitoring of turbidity [166], such as after heavy rainfall or runoff events.

GNDVI (Green Normalized Difference Vegetation Index): GNDVI enhances chlorophyll detection by focusing on the green reflectance peak and the strong NIR reflectance of healthy vegetation. The green band is responsive to chlorophyll levels, and the NIR band captures the overall health of vegetation. This sensitivity makes GNDVI effective for estimating nitrate concentrations and detecting early stages of algal blooms [166]. By using Sentinel-2 bands B8 (NIR) and B3 (green), GNDVI provides a more nuanced view of vegetation health compared to NDVI, making it useful in environments where early detection of vegetation stress is crucial.

EVI (Enhanced Vegetation Index): EVI improves upon NDVI by incorporating the blue band (Band 2) to correct for atmospheric scattering and soil background noise, making it more reliable under dense vegetation conditions. This index is particularly effective for monitoring dense algal blooms and assessing water body extents [167]. EVI's ability to reduce noise and improve signal quality makes it a robust tool for evaluating Chl-a concentrations and overall vegetation health in aquatic environments. The formula $2.5 \times (\text{NIR} - \text{Red})/(\text{NIR} + 6 \times \text{Red} - 7.5 \times \text{Blue} + 1)$ leverages the strengths of NIR, red, and blue bands to provide enhanced sensitivity.

NDWI (Normalized Difference Water Index): NDWI enhances open water detection by using the green (Band 3) and NIR (Band 8) bands. Water bodies reflect green light and absorb NIR light, creating a distinct contrast that NDWI exploits. This index is effective for monitoring changes in water extent and assessing changes in water extent, turbidity [166,168], and Chl-a levels [167]. NDWI's sensitivity to water content helps distinguish water from other land cover types [169], making it a significant index for accurate water quality monitoring.

NDCI (Normalized Difference Chlorophyll Index): NDCI utilizes the red-edge region, which is highly responsive to chlorophyll variations. The red-edge band captures subtle changes in chlorophyll content that other indices might miss, making NDCI particularly effective for assessing chlorophyll concentrations in turbid and nutrient-rich waters [170,172]. By using Sentinel-2 bands B5 (red edge 1) and B4 (red), NDCI provides enhanced sensitivity and accuracy in Chl-a estimation. This makes it a valuable tool for managing eutrophication and detecting algal blooms.

MCI (Maximum Chlorophyll Index): MCI focuses on the red-edge region to estimate high chlorophyll content. The red-edge bands capture the strong absorption feature of chlorophyll, allowing MCI to detect dense algal blooms effectively. This index is particularly useful for monitoring eutrophic lakes and identifying areas with excessive algal growth, providing precise detection of high chlorophyll levels [167].

NSMI (Normalized Suspended Material Index): NSMI estimates the concentration of suspended materials by contrasting red and shortwave infrared reflectance. Suspended particles affect the scattering properties of water, and NSMI leverages this characteristic to assess total suspended solids (TSS) [173]. The sensitivity of SWIR bands to particle scattering makes NSMI particularly effective for managing sediment load and water clarity, providing a robust measure of water quality.

SCI (Synthetic Chlorophyll Index): SCI is tailored for evaluating Chl-a concentrations in turbid and nutrient-rich waters. By contrasting NIR and red reflectance, SCI enhances the detection of chlorophyll. NIR and red bands are utilized for SCI, making it suitable for diverse aquatic environments particularly effective in areas where water is turbid, and traditional indices might struggle to provide accurate estimates.

The use of spectral indices in water quality assessment has demonstrated significant advantages, particularly in enhancing the detection and monitoring of key parameters across diverse aquatic environments. A review of various indices within this section reveals their capacity to capture essential water quality metrics, but the effectiveness of these indices can vary depending on the environmental context and specific parameter being analyzed. NDVI and NDWI, commonly employed for general assessments, have proven valuable in large-scale applications, yet their broad spectral approach may not always capture the nuances of more complex or turbid water bodies. Table 3 presents the spectral indices commonly used in lake water quality assessments.

Conversely, indices like NDCI and MCI, which leverage hyperspectral data, offer greater precision and are particularly effective in identifying subtle variations in parameters like Chl-a and other pigment concentrations.

The key challenge moving forward is refining these indices to better address the limitations associated with specific water quality conditions, such as high turbidity or mixed pixel effects. Integrating spectral indices with advanced machine learning algorithms presents a promising path forward, as this approach can enhance predictive accuracy and provide a more robust framework for real-time monitoring.

4. Advanced Machine Learning Methods for Water Quality Prediction

Machine learning models offer significant advantages by automatically learning patterns from large datasets without the need for explicit programming [174]. Machine learning models can also handle non-linear relationships and interactions between variables, leading to improved prediction accuracy and robustness [175]. These advancements enable more precise and scalable water quality assessments, making machine learning

methods essential for modern lake water quality monitoring and management. As shown in Figure 2, the primary categories of machine learning methods used include traditional machine learning models and deep learning models. Below, we detail the specific techniques within each category's specific techniques, mechanisms, advantages, breakthroughs, and challenges.

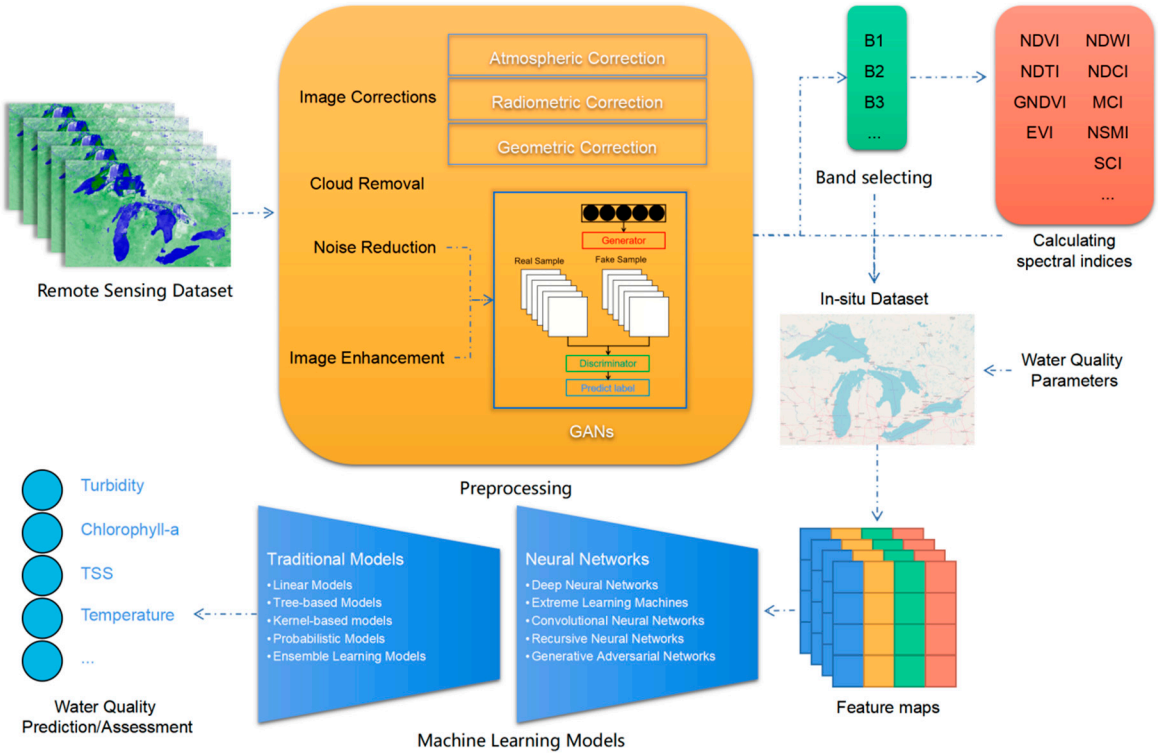


Figure 2. Workflow diagram of machine learning learning-based water quality parameters prediction using remote sensing dataset.

Traditional machine learning models are crucial and fundamental in predicting lake water quality using remote sensing data. Traditional machine learning models have long been applied to predict lake water quality based on remote sensing data. This section reviews studies from the past five years and finds that these traditional models continue to show significant promise in the prediction of lake water quality using remote sensing data.

Each model type offers unique advantages, and their application has been supported by various studies demonstrating their effectiveness in different contexts. Despite their early adoption, these models remain relevant and continue to play a crucial role in modern research. In this section, these models are categorized into several types: linear models, tree-based models, kernel-based models, probabilistic models, ensemble learning models, extreme learning machines (ELMs), convolutional neural networks (CNNs), recurrent neural networks (RNNs), and generative adversarial networks (GANs).

Linear models, like linear regression (LR) [176], ridge regression (RR) [177,178], and Lasso regression (LassoR) [178–180] are widely used for predicting lake water quality using remote sensing data due to their simplicity and quick assessment capabilities. These models presume a linear connection between features and the target, making them easy to implement and interpret, and ideal for initial analyses. Regularization methods in ridge and Lasso models improve their ability to handle complex data, enhancing robustness and stability. However, they often struggle with capturing non-linear relationships, limiting their effectiveness in more complex scenarios.

Tree-based models, including decision trees, RFs, and GBTs, are well-suited for predicting water quality from remote sensing data because of their capacity to capture complex, non-linear relationships. While decision trees are easy to interpret, they are often

overfit [168]; however, RFs mitigate this by averaging predictions across multiple trees, improving accuracy and robustness. Studies [177–180] have shown RFs effectively predict both optical (e.g., Chl-a) and non-optical (e.g., turbidity) variables with high interpretability. GBTs further enhance predictive performance by iteratively correcting errors, achieving high accuracy in challenging cases, as demonstrated by an R^2 of 0.90 in Chl-a mapping [155]. These models offer interpretability, accuracy, and resilience in handling the complexities of remote sensing data.

Kernel-based models like SVM excel in handling high-dimensional, non-linear data, making them ideal for remote sensing-based lake water quality prediction. SVM effectively classifies complex spectral patterns, as shown in studies [180] with high accuracy. For regression, SVR models non-linear relationships to estimate parameters like Chl-a, turbidity, and TSS, offering robust predictions with minimal error. Research [89] highlights SVR's efficiency in predicting water quality with fewer hyperparameters, making it a reliable choice for remote sensing applications.

Probabilistic models like naive Bayes and Gaussian process regression (GPR) excel in handling uncertainty, making them ideal for remote sensing water quality prediction. Naive Bayes efficiently classifies water quality categories, integrating prior data with observed inputs, as shown in algal bloom assessments using MODIS [181]. GPR, known for modeling complex relationships and providing confidence intervals, is particularly effective with small, noisy datasets. It outperforms empirical methods in predicting parameters like total phosphorus (TP) and chemical oxygen demand (COD) [52], offering more accurate and reliable estimates in environmental monitoring.

Ensemble learning models, such as bagging and boosting, combine multiple base models to enhance accuracy and robustness in remote sensing lake water quality prediction. Bagging techniques like RFs reduce variance and mitigate overfitting by averaging predictions from multiple decision trees trained on bootstrapped data subsets, making them effective for complex environmental datasets [177–180]. Boosting techniques, including GBTs, sequentially build models to correct previous errors, improving prediction accuracy. Studies have shown that advanced boosting methods like XGBoost, LightGBM, and CatBoost outperform single models in predicting critical water quality indicators including Chl-a, TP, and total nitrogen (TN) [182,183], demonstrating their superior capability in handling the complexities of remote sensing data.

Traditional machine learning models have proven their effectiveness in lake water quality prediction, particularly when dealing with structured data and capturing both linear and non-linear relationships. Their application in remote sensing has demonstrated improved accuracy over empirical methods, especially in scenarios where the data is well-defined and the relationships between variables are relatively straightforward. However, these models often require extensive feature engineering and may struggle with the complexity, variability, and high dimensionality inherent in large and diverse remote sensing datasets. While traditional models like linear regression (LR) and decision trees offer simplicity and interpretability, they can fall short in handling the intricate interactions present in high-dimensional spectral data. Ensemble learning models, such as RFs and GBTs, have mitigated some of these challenges by improving prediction accuracy and robustness, yet they still depend heavily on the quality and representativeness of the input features.

Moreover, the inherent uncertainty and noise in remote sensing data highlight the limitations of traditional models. Probabilistic models like naive Bayes and Gaussian process regression (GPR) have been instrumental in addressing these issues, providing a framework to incorporate uncertainty into predictions and offering more reliable estimates. However, the growing complexity and scale of remote sensing data necessitate even more advanced methodologies. As lake water quality monitoring increasingly relies on diverse and high-dimensional datasets, the limitations of traditional machine learning models become more apparent. This underscores the need for more advanced techniques, such as neural networks and deep learning, which can automatically extract features and learn complex data representations. These models offer enhanced predictive performance, scala-

bility, and the ability to adapt to the specific challenges of lake water quality monitoring, making them essential for future research and application in this field. Compared with traditional machine learning models, they are based on the neural structures of the human brain, which are designed to model complex patterns and interactions in large datasets, rendering them appropriate for lake water quality prediction using remote sensing data. Deep neural networks (DNNs) are composed of several layers of interconnected neurons that process inputs through weighted connections, making them effective at capturing the non-linear relationships often present in remote sensing data. DNNs, such as hybrid Bayesian probabilistic neural network [184] and MLP [89], have been successfully applied to predict water quality parameters like Chl-a and TSS, as demonstrated by studies [89,184] using hyperspectral and multispectral images.

ELMs simplify neural network training by randomly selecting input weights and biases, reducing computational costs while maintaining competitive performance. ELMs have shown high accuracy in predicting lake water quality indicators, effectively processing high-dimensional remote sensing data [57,185]. Their efficiency and robustness make ELMs a strong choice for handling complex spectral datasets, outperforming traditional techniques like band ratios and bio-optical inversion [185].

CNNs are built to process grid-like data, such as images, by employing convolutional layers that apply filters to input data, identifying local patterns and spatial hierarchies. This structure enables CNNs to effectively extract spatial features from satellite imagery, making them ideal for water quality prediction. Pooling layers further reduce data dimensionality while retaining crucial features, with activation functions like ReLU adding non-linearity, allowing the network to learn complex patterns [175]. CNNs' robustness in handling complex data and anomalies, with models like AlexNet and ResNet50 showing superior performance in classifying water quality levels [186–188].

RNNs are built to handle sequential data by retaining a hidden state that stores information from prior inputs, making them particularly effective for temporal tasks. RNNs and their variants, such as long short-term memory (LSTM) [189,190], ConvLSTM [191,192], CNN–LSTM [193] and GRU [194], excel at integrating spatial and temporal features, making them well-suited for dynamic water quality prediction. RNN-based models effectively capture temporal dependencies in remote sensing data, improving prediction accuracy for parameters like particulate organic carbon and lake surface water temperature [195,196]. These models are particularly valuable for addressing the complexities of spatiotemporal environmental monitoring.

GANs consist of two neural networks—a generator and a discriminator—that are trained simultaneously through an adversarial process. The generator produces synthetic images, while the discriminator assesses their authenticity by differentiating them from real images. This adversarial training enables the generator to progressively enhance the resolution and quality of its outputs. GANs are particularly effective in remote sensing based water quality prediction. For instance, self-attention GANs have been employed to focus on significant image areas, improving the detection of algae blooms [197]. Additionally, enhanced models like Pix2Pix GAN have shown superior performance in correcting distortions in remote sensing images, providing accurate predictions for water quality parameters in smart city environments [198,199].

Table 4 outlines the train-test splits and the volume of ground truth data utilized in the corresponding studies cited. For cases where the total amount of data was not explicitly mentioned, we estimated the data volume under the assumption that the remote sensing images covered all monitoring stations. The results of this analysis are summarized in Table 4.

Table 4. Machine Learning Models in Water Quality Assessment.

Model	Train: Test	Number of Data	Ref.
LR	3:1	156	[176]
RR	4:1	30	[177]
LassoR	8:3	112	[179]
RFs	4:1	30	[177]
GBTs	2:1	329	[155]
SVM	8:3	112	[179]
SVR	4:1	382	[89]
GPR	2:1	374	[52]
DNNs	4:1	382	[89]
CNN	4:1	465	[186]
ConvLSTM	4:1	8744	[191]
CNN-LSTM	2:1	76	[193]
GANs	4:1	898	[197]

The "Number of Data" column refers to the number of ground truth data points used in these studies for training and validating the models.

One of the most critical considerations when choosing a machine learning model for lake water quality prediction is the balance between complexity and interpretability. Simpler models, including LR and decision trees, offer clear and interpretable results, making them appealing for scenarios where transparency is essential, such as policy-making or public communication [200]. However, these models often fall short when tasked with capturing the non-linear relationships inherent in environmental data [200]. As water quality parameters like Chl-a, turbidity, and TSS exhibit complex interactions influenced by numerous factors, more sophisticated models like RFs and SVMs become necessary to improve predictive accuracy.

The effectiveness of machine learning models is heavily dependent on the availability and quality of ground truth data. As shown in Table 4, linear models [176,177,179] require relatively modest amounts of data to function adequately, but their predictions can be overly simplistic and less reliable for complex systems. By contrast, deep learning models, including CNNs [186] and LSTM networks [191–193], demand large, high-quality datasets to reach their full potential. The trade-off here is clear: while more complex models can offer superior performance, they also require significantly more data and computational resources. For example, studies leveraging CNNs for spatially complex parameters like algal blooms demonstrate impressive predictive capabilities, but at the cost of extensive training data and computational power [184,187].

Different water quality parameters present unique challenges that influence the choice of machine learning models. For optical active parameters like Chl-a and turbidity, models that can capture spectral variations effectively, such as RFs and GBTs, have shown strong performance [177–180]. These models are particularly adept at handling the variability in spectral reflectance associated with these parameters. For non-optical parameters or those with more subtle spectral signatures, such as CDOM and TSS, kernel-based models like SVMs and ensemble methods tend to offer better results, given their ability to manage data with high dimensionality and intricate relationships.

The temporal and spatial resolution of the satellite data is another crucial factor that dictates model choice. For instance, satellites with higher temporal resolution, like MODIS, are more suitable for monitoring rapidly changing parameters, such as dissolved oxygen or algal blooms, where timely observations are essential [43,149–152]. However, MODIS's lower spatial resolution limits its effectiveness in nearshore pollution detection, where fine-scale spatial detail is essential. Conversely, Sentinel-2's higher spatial resolution makes it better suited for pinpointing pollution sources and monitoring smaller, more sensitive water bodies, but its longer revisit time may miss short-term changes. This emphasizes the need for models that can integrate both temporal and spatial dynamics, such as CNN–LSTM hybrids, which capture spatial features while accounting for temporal variations [191–193].

Overfitting remains a perennial challenge, particularly for more complex models like neural networks and ensemble methods. While these models excel in capturing detailed patterns within training data, they risk failing to generalize well to new, unseen data if not properly regularized [201]. Techniques such as cross-validation, dropout in neural networks, and the use of ensemble models help mitigate this risk [201]. However, achieving strong generalization requires a careful balance, which largely depends on the quality and diversity of the training data. Studies that employ methods like RFs have shown that while these models are generally resistant to overfitting, they still require careful tuning and sufficient data diversity to maintain predictive accuracy across different lake environments [177–180].

The complexity of lake ecosystems and the variability of water quality parameters often necessitate the integration of multiple models. Hybrid approaches that integrate the advantages of various algorithms, like CNNs for spatial feature extraction and LSTM for temporal sequence modeling, have shown enhanced performance in capturing the complex aspects of lake water quality [191–193].

Similarly, the use of GANs to enhance data quality before applying more traditional predictive models can improve overall accuracy by reducing noise and filling in data gaps [197,198]. The integration of models is particularly valuable in cases where different parameters require different modeling approaches, or when data availability is uneven across the spatial and temporal dimensions.

The choice of machine learning model for lake water quality prediction must be carefully aligned with the specific objectives of the monitoring program, the characteristics of the lake, the available data, and the size of the dataset. The selection of the appropriate machine learning model for water quality prediction depends on several factors, including the complexity of the water body's optical properties, the type of water quality parameters being measured, and the availability and quality of data [202]. In optically clear waters, simpler models, such as linear regression [176] or SVM [179], may suffice for parameters like chlorophyll-a. However, in optically complex environments, such as lakes with high turbidity or dense vegetation, more robust models like RF, GBTs [177–180], or DNN are better suited due to their ability to handle non-linear relationships and large datasets [179,203]. For lakes with highly variable seasonal dynamics, LSTM or RNNs can be used to model temporal changes and improve prediction accuracy [191–193].

More advanced models, like RFs and GBTs, offer greater accuracy and flexibility, especially with Landsat data, where they effectively handle moderate-resolution data and mitigate overfitting, making them particularly suited for parameters like turbidity [179,203]. These models are robust even when the dataset size is moderate, balancing complexity and performance.

SVM [179] and SVR [89] models excel in extracting meaningful patterns from high-resolution spectral data, making them well-suited for Landsat datasets where high-dimensional and non-linear relationships are prevalent. However, they need a significant volume of data to perform optimally, particularly in capturing subtle variations in water quality parameters. For high-resolution datasets, like those from Sentinel-2, CNNs have demonstrated strong performance in spatial feature extraction, allowing for detailed analyses of water quality parameters [186]. CNNs are particularly effective when large datasets are available, as they can leverage extensive data to improve model accuracy.

Although computationally intensive, GANs are emerging as powerful tools for enhancing and correcting remote sensing images, making them ideal for use with hyperspectral data to improve the resolution and clarity of predictions in large-scale monitoring efforts [197]. However, their effectiveness is closely tied to the availability of large training datasets, which are necessary to optimize their performance and reduce artifacts in generated outputs.

The key to successful application lies in selecting the appropriate model (or combination of models) that balances these trade-offs, ensuring both robustness and practical relevance for ongoing environmental monitoring efforts. Future research should continue

to explore hybrid models and data augmentation techniques, such as combining CNNs for spatial analysis with LSTM for temporal dynamics, to optimize this balance.

By capitalizing on the advantages of different models and taking dataset size into account, researchers can propel the field of remote sensing and machine learning for water quality management, resulting in more accurate and reliable predictions that enhance decision-making for lake ecosystems.

5. Integration of Machine Learning and Remote Sensing

Traditional approaches, which rely on manual sampling and in situ measurements, are limited by their frequency and coverage. The combination of remote sensing and machine learning offers a transformative solution, enabling continuous, large-scale water quality assessment and proactive management. This review examines how the integration of these technologies advances water quality management by improving decision-making, enhancing early warning systems, and optimizing resource allocation strategies.

The integration of machine learning with remote sensing has significantly improved the capacity of early warning systems (EWS) to detect and predict harmful events, such as harmful algal blooms (HABs) and pollution spikes. Traditional methods often respond too late, after environmental damage has already occurred. By contrast, ML-powered systems can process real-time satellite data and provide early warnings, allowing for timely intervention. In Taihu Lake, Xu [204] showed that SVMs could detect early signs of water quality degradation from satellite imagery, offering lead time for management actions to mitigate potential HABs. This approach illustrates how machine learning, coupled with remote sensing, can move water quality management from reactive to proactive, preventing the escalation of environmental issues. Hill et al. [205] discussed the use of machine learning architectures, including CNNs, SVMs, and RFs, integrated with MODIS satellite data to detect and predict harmful algal bloom events, specifically Karenia brevis in Florida's coastal waters. The model achieved a detection accuracy of 91% and predicted HABs up to 8 days in advance with 86% accuracy, demonstrating the effectiveness of machine learning in monitoring and forecasting HAB occurrences.

Additionally, Chang et al. [206] demonstrated the value of combining data from multiple satellites, such as Landsat and Sentinel, with machine learning to improve prediction accuracy. By merging multisensor data, the study enhanced temporal and spatial coverage, allowing for continuous monitoring even when cloud cover or sensor limitations would otherwise disrupt data collection. This capability is essential for maintaining a comprehensive and up-to-date understanding of water quality trends. Sagan et al. [57] applied DNNs, LSTM to hyperspectral remote sensing data, successfully predicting key water quality indicators such as chlorophyll-a and turbidity. The study emphasized the potential of machine learning for anomaly detection and pollution forecasting, demonstrating its applicability for managing water quality and preventing pollution spikes.

Beyond prediction and monitoring, machine learning plays a crucial role in optimizing water quality management strategies. By simulating different management scenarios, machine learning models help decision-makers evaluate the potential outcomes of various interventions, such as nutrient load reduction or water treatment enhancements, before implementing them in real life. Sedighkia et al. [207] utilized reinforcement learning models integrated with satellite data to simulate nutrient reduction strategies in lakes. This approach helped identify the most effective methods for controlling eutrophication, demonstrating that ML not only enhances prediction accuracy but informs resource-efficient decision-making. Simulating multiple intervention scenarios ensures that managers can allocate resources to interventions that yield the best environmental outcomes.

Moreover, real-time adaptive management is becoming increasingly feasible with the integration of machine learning and remote sensing. Li et al. [208] demonstrated how machine learning algorithms such as RFs, SVR, and ANN, when applied to hyperspectral satellite data, allowed for continuous monitoring and dynamic adjustments to management strategies in Lake Ontario. As conditions evolved, the model adjusted recommendations in

real time, ensuring that resources were directed to the areas where they were most needed, optimizing both environmental and economic outcomes.

The integration of machine learning with remote sensing significantly enhances lake water quality management. It provides timely early warnings and optimize resource use. By improving both monitoring and strategic planning, machine learning driven remote sensing offers a proactive and efficient approach to addressing water quality challenges.

6. Challenges and Future Direction of Research

The use of remote sensing and machine learning in lake water quality prediction faces significant challenges, particularly in data quality, model development, and practical implementation. Data quality is a major issue, as high-resolution, cloud-free images are often compromised by atmospheric conditions like cloud cover and haze, leading to inaccuracies. The temporal resolution of satellite data may not align with monitoring needs, and inconsistencies in spatial resolution across platforms complicate data integration [120]. To address these issues, multi-sensor data fusion, advanced machine learning techniques for cloud removal and image enhancement, and the use of drones and ground-based sensors can improve data reliability and coverage [57]. Standardized data preprocessing protocols are also essential for consistency.

In model development, challenges include overfitting, underfitting, and the complexity of hyperparameter tuning. Overfitting occurs when models capture noise rather than meaningful patterns, while underfitting results from overly simplistic models. Additionally, advanced models like deep learning often function as "black boxes" lacking interpretability [209]. To mitigate these issues, robust cross-validation, regularization techniques, automated hyperparameter optimization, and model interpretability tools like SHAP and LIME are essential. Combining different algorithms in model ensembles can also enhance robustness [210].

Practical implementation of these models involves challenges such as the need for substantial computational resources, multi-source data integration, and translating complex outputs into actionable insights for stakeholders. Ensuring data privacy and security is also critical. Cloud-based platforms and high-performance computing can meet computational demands, while standardized frameworks and user-friendly decision support systems can facilitate data integration and application. Robust data governance policies are necessary to maintain privacy and security throughout the process [211].

Overcoming these challenges in data quality, model development, and implementation is essential for advancing the application of remote sensing and machine learning in lake water quality management. Through improved data integration, model accuracy, interpretability, and robust implementation, these technologies can provide reliable and actionable insights for environmental monitoring and management.

As the field of remote sensing and machine learning continues to evolve, its application in lake water quality management holds significant potential. To fully leverage these technologies, future research must focus on developing more robust, adaptable models capable of addressing the diverse challenges posed by different environmental conditions. Key advancements will include the application of transfer learning to improve model generalizability across various datasets, thus enabling more reliable predictions under varying conditions [212].

The integration of advanced data sources—such as high-frequency in situ sensors, autonomous monitoring drones, and IoT networks—will be essential. These technologies provide richer, real-time datasets with finer spatial and temporal resolution, significantly improving the accuracy and reliability of water quality predictions. Combining these data sources with traditional remote sensing techniques will result in more comprehensive and effective monitoring systems.

Furthermore, improving data fusion techniques, especially deep learning-based methods that blend multispectral, hyperspectral, and radar data, will yield more precise water quality assessments. Emerging machine learning approaches, such as reinforcement learn-

ing and federated learning, offer additional opportunities to revolutionize monitoring strategies and collaborative model training, while maintaining data privacy and improving prediction accuracy.

7. Conclusions

This review has provided an in-depth analysis of the integration of remote sensing and machine learning for lake water quality management, focusing on the suitability of different satellites, the prediction of various water quality parameters, and the effectiveness of different machine learning models in utilizing satellite data. The findings offer critical insights into how these technologies can be optimized for specific environmental monitoring tasks.

This review identified that different satellites are better suited for different types of lakes and specific water quality parameters. For instance, Landsat satellites (e.g., Landsat 5, 7, and 8) with their long operational history and moderate spatial resolution are ideal for large lakes where long-term trend analysis is critical. These satellites are particularly effective for monitoring parameters like Chl-a [155] and temperature [97]. On the other hand, Sentinel-2 offers higher spatial resolution and more frequent revisit times, making it suitable for medium-sized lakes and more dynamic parameters such as CDOM [137,141] and TSS [139]. MODIS, with its broad spectral coverage and frequent revisits, is highly effective for very large lakes and for parameters that require frequent monitoring, such as surface temperature [133–136] and algal blooms [43,149–152]. RapidEye, with its fine spatial resolution, is well-suited for small lakes or specific localized studies [79,145,149].

Different water quality parameters require specific combinations of satellites and predictive models for optimal accuracy. For Chl-a, Landsat paired with models like RFs [79] have shown high predictive accuracy due to their ability to capture spectral variations effectively. Turbidity is well predicted using Sentinel-2 data [87,89] in conjunction with GBTs [87] and MLP [89], which can handle the complex relationships in the spectral data. For CDOM, MODIS [95], and Sentinel-2 [94] datasets offer superior performance due to the high temporal or spatial resolution.

This review also analyzed which machine learning models are best suited for different satellite datasets and ground truth dataset sizes. RFs and GBTs are highly effective with Landsat data [179,203], especially for parameters like turbidity, due to their robustness in handling moderate-resolution data and their ability to reduce overfitting. These models perform well even when the ground truth dataset size is moderate, striking a balance between complexity and performance. SVM and SVR are particularly suited for Landsat data, where high-dimensional and non-linear relationships are prevalent. However, they require substantial ground truth data to fully capture the complexities of water quality parameters, making them more effective when larger datasets are available. CNNs, with their strong performance in spatial feature extraction, are well-matched with high-resolution datasets from Sentinel-2 for detailed analyses [186]. These models also benefit from larger ground truth datasets, which allow them to leverage extensive spatial data for improved accuracy. GANs, though computationally intensive, are emerging as powerful tools for enhancing and correcting remote sensing images. They are particularly well-suited for hyperspectral data, where they can improve resolution and clarity in large-scale monitoring efforts [197]. However, the effectiveness of GANs is closely tied to the availability of large training datasets, which are necessary to optimize their performance and reduce artifacts.

Interdisciplinary collaboration among environmental scientists, data scientists, and policymakers will be crucial in translating these technological advancements into practical management strategies. By incorporating scientific insights into policy frameworks and engaging stakeholders in co-developing solutions, the effective implementation of these technologies can be significantly enhanced, ultimately leading to improved protection and management of lake ecosystems.

This review highlights the importance of selecting the right combination of satellite platforms, water quality parameters, and machine learning models to optimize lake water

quality monitoring. While advanced models like CNNs, RNNs, and GANs offer high accuracy, their applicability depends on the quality and resolution of the satellite data. Future research should focus on refining these combinations and addressing challenges in data integration, model robustness, and computational efficiency to fully leverage the potential of these technologies in environmental management.

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References

1. Shivaraju, V.M. Faunal Diversity of Durgadahalli Lake of Tumakuru, Karnataka State, India. *J. Environ. Stud.* **2021**, *7*, 6. [CrossRef]
2. Cai, Y.; Dong, R.; Kattel, G.; Zhang, Y.; Peng, K.; Gong, Z. Macroinvertebrate diversity and ecosystem functioning across the eutrophication gradients of the middle and lower reaches of Yangtze River lakes (China). *Ecol. Evol.* **2023**, *13*, e9751. [CrossRef] [PubMed]
3. The Canadian Encyclopedia. Geography of Ontario. Available online: https://www.thecanadianencyclopedia.ca/en/article/geography-of-ontario (accessed on 30 June 2024).
4. Great Lakes Fishery Commission. The Great Lakes Fishery. Available online: https://www.glfc.org/the-fishery.php (accessed on 25 June 2024).
5. Schirpke, U.; Scolozzi, R.; Tappeiner, U. Not too small to benefit society: Insights into perceived cultural ecosystem services of mountain lakes in the European Alps. *Ecol. Soc.* **2022**, *7*, 1. [CrossRef]
6. Jabbari, A.; Ackerman, J.; Boegman, L.; Zhao, Y. Increases in Great Lake winds and extreme events facilitate interbasin coupling and reduce water quality in Lake Erie. *Sci. Rep.* **2021**, *11*, 5733. [CrossRef] [PubMed]
7. Shrestha, N.; Seglenieks, F.; Temgoua, A.; Dehghan, A. The Impacts of Climate Change on Land Hydroclimatology of the Laurentian Great Lakes Basin. *Front. Water* **2022**, *4*, 801134. [CrossRef]
8. Wang, Z.; Zhang, T.; Tan, C.; Xue, L.; Bukovsky, M.; Qi, Z. Modeling impacts of climate change on crop yield and phosphorus loss in a subsurface drained field of Lake Erie region, Canada. *Agric. Syst.* **2021**, *190*, 103110. [CrossRef]
9. Austin, J.C. *America's North Coast: A Benefit-Cost Analysis of a Program to Protect and Restore the Great Lakes*; Brookings Institution: Washington, DC, USA, 2007.
10. Jiang, B.; Mitsch, W.; Lenhart, C. Estimating the Importance of Hydrologic Conditions on Nutrient Retention and Plant Richness in a Wetlaculture Mesocosm Experiment in a Former Lake Erie Basin Swamp. *Water* **2021**, *13*, 2509. [CrossRef]
11. Smith, R.B.; Bass, B.; Sawyer, D.; Depew, D.; Watson, S.B. Estimating the economic costs of algal blooms in the Canadian Lake Erie Basin. *Harmful Algae* **2019**, *87*, 101624. [CrossRef] [PubMed]
12. Schindler, D.W. Lakes as sentinels and integrators for the effects of climate change on watersheds, airsheds, and landscapes. *Limnol. Oceanogr.* **2009**, *54 Pt 2*, 2349–2358. [CrossRef]
13. Vörösmarty, C.J.; Green, P.; Salisbury, J.; Lammers, R.B. Global water resources: Vulnerability from climate change and population growth. *Science* **2000**, *289*, 284–288. [CrossRef]
14. O'Reilly, C.M.; Sharma, S.; Gray, D.K.; Hampton, S.E.; Read, J.S.; Rowley, R.J.; Schneider, P.; Lenters, J.D.; McIntyre, P.B.; Kraemer, B.M.; et al. Rapid and highly variable warming of lake surface waters around the globe. *Geophys. Res. Lett.* **2015**, *42*, 10773–10781. [CrossRef]
15. Hamilton, D.P.; Mitchell, S.F. Wave-induced shear stresses, plant nutrients and chlorophyll in seven shallow lakes. *Freshw. Biol.* **1997**, *38*, 159–168. [CrossRef]
16. Wetzel, R.G. *Limnology: Lake and River Ecosystems*; Gulf Professional Publishing: Houston, TX, USA, 2001.
17. Cloern, J.E. Our evolving conceptual model of the coastal eutrophication problem. *Mar. Ecol. Prog. Ser.* **2001**, *210*, 223–253. [CrossRef]
18. Ruhl, H.A.; Rybicki, N.B. Long-term reductions in anthropogenic nutrients link to improvements in Chesapeake Bay habitat. *Proc. Natl. Acad. Sci. USA* **2010**, *107*, 16566–16570. [CrossRef] [PubMed]
19. Venkateswaran, P.; Suresh, M.A.; Venkatasubramanian, N. Augmenting In-Situ with Mobile Sensing for Adaptive Monitoring of Water Distribution Networks. In Proceedings of the 10th ACM/IEEE International Conference on Cyber-Physical Systems, Montreal, QC, Canada, 16–18 April 2019; pp. 151–162.

20. Pu, F.; Ding, C.; Chao, Z.; Yu, Y.; Xu, X. Water-quality classification of inland lakes using Landsat8 images by convolutional neural networks. *Remote Sens.* **2019**, *11*, 1674. [CrossRef]
21. Schindler, D.W. Whole-ecosystem experiments: Replication versus realism: The need for ecosystem-scale experiments. *Ecosystems* **1998**, *1*, 323–334. [CrossRef]
22. Ellison, R.; Slocum, D. High Spatial Resolution Mapping of Water Quality and Bathymetry with a Person-Deployable, Low-Cost Autonomous Underwater Vehicle. In Proceedings of the OCEANS, Quebec City, QC, Canada, 15–18 September 2008; pp. 1–7.
23. Babitsch, D.; Berger, E.; Sundermann, A. Linking environmental with biological data: Low sampling frequencies of chemical pollutants and nutrients in rivers reduce the reliability of model results. *Sci. Total Environ.* **2021**, *772*, 145498. [CrossRef]
24. Tucker, C.J. Red and photographic infrared linear combinations for monitoring vegetation. *IEEE Trans. Geosci. Remote Sens.* **1979**, *GE-17*, 126–144. [CrossRef]
25. Hansen, M.C.; Potapov, P.V.; Moore, R.; Hancher, M.; Turubanova, S.A.; Tyukavina, A.; Thau, D.; Stehman, S.V.; Goetz, S.J.; Loveland, T.R.; et al. High-resolution global maps of 21st-century forest cover change. *Science* **2013**, *342*, 850–853. [CrossRef]
26. Middel, A.; Lukasczyk, J.; Zakrzewski, S.; Arnold, M.; Maciejewski, R. Urban form and composition of street canyons: A human-centric big data and deep learning approach. *Landsc. Urban Plan.* **2019**, *183*, 122–132. [CrossRef]
27. Stumpf, R.P.; Wynne, T.T.; Baker, D.B.; Fahnenstiel, G.L. Interannual variability of cyanobacterial blooms in Lake Erie. *PLoS ONE* **2012**, *7*, e42444. [CrossRef] [PubMed]
28. Dekker, A.G.; Peters, S.W. The use of the thematic mapper for the analysis of eutrophic lakes: A case study in the Netherlands. *Int. J. Remote Sens.* **1993**, *14*, 799–821. [CrossRef]
29. Gitelson, A.A.; Kondratyev, K.Y. Optical models of mesotrophic and eutrophic water bodies. *Int. J. Remote Sens.* **1991**, *12*, 603–615. [CrossRef]
30. Kutser, T.; Pierson, D.C.; Kallio, K.Y.; Reinart, A.; Sobek, S. Mapping lake dissolved organic matter (DOM) by satellite remote sensing. *Remote Sens. Environ.* **2005**, *94*, 535–540. [CrossRef]
31. Olmanson, L.G.; Bauer, M.E.; Brezonik, P.L. A 20-year Landsat water clarity census of Minnesota's 10,000 lakes. *Remote Sens. Environ.* **2008**, *112*, 4086–4097. [CrossRef]
32. Thoreson, B.D. Remote Sensing of Water Quality in Prairie Lakes. Master's Thesis, South Dakota State University, Brookings, SD, USA, 1975.
33. Alföldi, T.T.; Munday, J.C., Jr. Water quality analysis by digital chromaticity mapping of Landsat data. *Can. J. Remote Sens.* **1978**, *4*, 108–126. [CrossRef]
34. Hergenrader, G.L. *Application of Remote Sensing in the Determination of Water Quality in Nebraska Reservoirs*; NASA Contractor Report No. NASA-CR-148776; NASA: Washington, DC, USA, 1976.
35. Zhang, B.; Zhang, B.; Hong, M.; Duan, H.T.; Song, K.S.; Wang, Z.M. Advance in remote sensing of lake water quality. *Adv. Water Sci.* **2007**, *18*, 301–310.
36. Ritchie, J.; Zimba, P.; Everitt, J. Remote Sensing Techniques to Assess Water Quality. *Photogramm. Eng. Remote Sens.* **2003**, *69*, 695–704. [CrossRef]
37. Odermatt, D.; Gitelson, A.; Brando, V.; Schaepman, M. Review of constituent retrieval in optically deep and complex waters from satellite imagery. *Remote Sens. Environ.* **2012**, *118*, 116–126. [CrossRef]
38. Härmä, P.; Vepsäläinen, J.; Hannonen, T.; Pyhälahti, T.; Kämäri, J.; Kallio, K.; Eloheimo, K.; Koponen, S. Detection of water quality using simulated satellite data and semi-empirical algorithms in Finland. *Sci. Total Environ.* **2001**, *268*, 107–121. [CrossRef]
39. Doerffer, R.; Sørensen, K.; Aiken, J. MERIS potential for coastal zone applications. *Int. J. Remote Sens.* **1999**, *20*, 1809–1818. [CrossRef]
40. Giardino, C.; Bresciani, M.; Villa, P.; Martinelli, A. Application of remote sensing in water resource management: The case study of Lake Trasimeno, Italy. *Water Resour. Manag.* **2010**, *24*, 3885–3899. [CrossRef]
41. Hoogenboom, H.; Dekker, A. Simulation of the medium-resolution imaging spectrometer (MERIS) performance for detecting chlorophyll-a over turbid inland waters. *SPIE* **1997**, *2963*, 440–447. [CrossRef]
42. Shen, F.; Verhoef, W. Suppression of local haze variations in MERIS images over turbid coastal waters for retrieval of suspended sediment concentration. *Opt. Express* **2010**, *18*, 12653–12662. [CrossRef]
43. Zhu, L.; Wang, S.; Zhou, Y.; Yan, F.; Zhou, W. Water Quality Monitoring in Taihu Lake Using MODIS Image Data. In Proceedings of the IGARSS 2004 IEEE International Geoscience and Remote Sensing Symposium, Anchorage, AK, USA, 20–24 September 2004; Volume 4, pp. 2314–2317. [CrossRef]
44. Gitelson, A.A.; Dall'Olmo, G.; Moses, W.; Rundquist, D.C.; Barrow, T.; Fisher, T.R.; Holz, J. A simple semi-analytical model for remote estimation of chlorophyll-a in turbid waters: Validation. *Remote Sens. Environ.* **2008**, *112*, 3582–3593. [CrossRef]
45. Ammenberg, P.; Flink, P.; Lindell, T.; Pierson, D.; Strombeck, N. Bio-optical modelling combined with remote sensing to assess water quality. *Int. J. Remote Sens.* **2002**, *23*, 1621–1638. [CrossRef]
46. Pierson, D.; Strömbeck, N. Estimation of radiance reflectance and the concentrations of optically active substances in Lake Mälaren, Sweden, based on direct and inverse solutions of a simple model. *Sci. Total Environ.* **2001**, *268*, 171–188. [CrossRef]
47. Hinton, G.; Deng, L.; Yu, D.; Dahl, G.E.; Mohamed, A.R.; Jaitly, N.; Kingsbury, B. Deep neural networks for acoustic modeling in speech recognition: The shared views of four research groups. *IEEE Signal Process. Mag.* **2012**, *29*, 82–97. [CrossRef]
48. Schmidhuber, J. Deep learning in neural networks: An overview. *Neural Netw.* **2015**, *61*, 85–117. [CrossRef]

49. Lu, S.; Tian, Y.Q.; Zhou, Q.; Shi, K. Remote Sensing Estimation of Chlorophyll-a Concentration for Optically Complex Waters Using Machine Learning Methods—A Case Study of Lake Erie, USA. *IEEE Trans. Geosci. Remote Sens.* **2019**, *57*, 8984–8996.
50. Cao, Z.; Ma, R.; Duan, H.; Pahlevan, N.; Melack, J.; Shen, M.; Xue, K. A machine learning approach to estimate chlorophyll-a from Landsat-8 measurements in inland lakes. *Remote Sens. Environ.* **2020**, *248*, 111974. [CrossRef]
51. Cao, Y.; Ye, Y.; Zhao, H.; Jiang, Y.; Wang, H.; Shang, Y.; Wang, J. Remote sensing of water quality based on HJ-1A HSI imagery. *Ecol. Inform.* **2018**, *44*, 21–32. [CrossRef]
52. Sun, X.; Zhang, Y.; Shi, K.; Zhang, Y.; Li, N.; Wang, W.; Huang, X.; Qin, B. Monitoring water quality using proximal remote sensing technology. *Sci. Total Environ.* **2021**, *803*, 149805. [CrossRef] [PubMed]
53. Wiangwang, N.; Qi, J. Water Quality Assessment Using Hyperspectral Remote Sensing. In Proceedings of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS '05), Seoul, Republic of Korea, 29 July 2005; Volume 7, pp. 4531–4534. [CrossRef]
54. Bresciani, M.; Giardino, C.; Fabbretto, A.; Pellegrino, A.; Mangano, S.; Free, G.; Pinardi, M. Application of new hyperspectral sensors in the remote sensing of aquatic ecosystem health. *Resources* **2022**, *11*, 8. [CrossRef]
55. Zhang, L.; Zhang, L. Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities. *IEEE Geosci. Remote Sens. Mag.* **2022**, *10*, 270–294. [CrossRef]
56. Ball, J.E.; Anderson, D.T.; Chan, C.S. Comprehensive survey of deep learning in remote sensing: Theories, tools, and challenges for the community. *J. Appl. Remote Sens.* **2017**, *11*, 042609. [CrossRef]
57. Sagan, V.; Peterson, K.T.; Maimaitijiang, M.; Sidike, P.; Sloan, J.; Greeling, B.A.; Maalouf, S.; Adams, C. Monitoring inland water quality using remote sensing: Potential and limitations of spectral indices, bio-optical simulations, machine learning, and cloud computing. *Earth-Sci. Rev.* **2020**, *205*, 103187. [CrossRef]
58. Wu, J.; Huang, X.; Xu, N.; Zhu, Q.; Zorn, C.; Guo, W.; Yu, C. Combining Satellite Imagery and a Deep Learning Algorithm to Retrieve the Water Levels of Small Reservoirs. *Remote Sens.* **2023**, *15*, 5740. [CrossRef]
59. Wang, X.; Li, Y.; Qiao, Q.; Tavares, A.; Liang, Y. Water quality prediction based on machine learning and comprehensive weighting methods. *Entropy* **2023**, *25*, 1186. [CrossRef]
60. Schindler, D.W. The cumulative effects of climate warming and other human stresses on Canadian freshwaters in the new millennium. *Can. J. Fish. Aquat. Sci.* **2001**, *58*, 18–29. [CrossRef]
61. Vörösmarty, C.J.; McIntyre, P.B.; Gessner, M.O.; Dudgeon, D.; Prusevich, A.; Green, P.; Davies, P.M. Global threats to human water security and river biodiversity. *Nature* **2010**, *467*, 555–561. [CrossRef] [PubMed]
62. Xiong, J.; Lin, C.; Ma, R.; Cao, Z. Remote Sensing Estimation of Lake Total Phosphorus Concentration Based on MODIS: A Case Study of Lake Hongze. *Remote Sens.* **2019**, *11*, 2068. [CrossRef]
63. Dong, L.; Gong, C.; Wang, X.; Wang, Y.; He, D.; Hu, Y.; Li, L.; Yang, Z. Seasonal Monitoring Method for TN and TP Based on Airborne Hyperspectral Remote Sensing Images. *Remote Sens.* **2024**, *16*, 1614. [CrossRef]
64. Zhang, L.; Zhang, L.; Cen, Y.; Wang, S.; Zhang, Y.; Huang, Y.; Sultan, M.; Tong, Q. Prediction of Total Phosphorus Concentration in Macrophytic Lakes Using Chlorophyll-Sensitive Bands: A Case Study of Lake Baiyangdian. *Remote Sens.* **2022**, *14*, 3077. [CrossRef]
65. Guo, H.; Huang, J.J.; Zhu, X.; Tian, S.; Wang, B. Spatiotemporal variation reconstruction of total phosphorus in the Great Lakes since 2002 using remote sensing and deep neural network. *Water Res.* **2024**, *255*, 121493. [CrossRef]
66. Yu, Z.; Yang, K.; Luo, Y.; Shang, C.; Zhu, Y. Lake surface water temperature prediction and changing characteristics analysis-A case study of 11 natural lakes in Yunnan-Guizhou Plateau. *J. Clean. Prod.* **2020**, *276*, 122689. [CrossRef]
67. Guo, Y.; Deng, R.; Li, J.; Hua, Z.; Wang, J.; Zhang, R.; Liang, Y.; Tang, Y. Remote sensing retrieval of total nitrogen in the Pearl River Delta based on Landsat8. *Water* **2022**, *14*, 3710. [CrossRef]
68. Wang, J.; Shi, T.; Yu, D.; Teng, D.; Ge, X.; Zhang, Z.; Yang, X.; Wang, H.; Wu, G. Ensemble machine-learning-based framework for estimating total nitrogen concentration in water using drone-borne hyperspectral imagery of emergent plants: A case study in an arid oasis, NW China. *Environ. Pollut.* **2020**, *266*, 115412. [CrossRef]
69. Japitana, M.; Burce, M.E. A Satellite-based Remote Sensing Technique for Surface Water Quality Estimation. *Eng. Technol. Appl. Sci. Res.* **2019**, *9*, 3965–3970. [CrossRef]
70. Batur, E.; Maktav, D. Assessment of Surface Water Quality by Using Satellite Images Fusion Based on PCA Method in the Lake Gala, Turkey. *IEEE Trans. Geosci. Remote Sens.* **2018**, *57*, 2983–2989. [CrossRef]
71. Pereira, O.J.R.; Merino, E.R.; Montes, C.; Barbiero, L.; Rezende-Filho, A.T.; Lucas, Y.; Melfi, A.J. Remote sensing Estimating Water pH Using Cloud-Based Landsat Images for a New Classification of the Nhecolândia Lakes (Brazilian Pantanal). *Remote Sens.* **2020**, *12*, 1090. [CrossRef]
72. Yin, Z.; Li, J.; Zhang, B.; Liu, Y.; Yan, K.; Gao, M.; Xie, Y.; Zhang, F.; Wang, S. Increase in chlorophyll-a concentration in Lake Taihu from 1984 to 2021 based on Landsat observations. *Sci. Total Environ.* **2023**, *873*, 162168. [CrossRef] [PubMed]
73. Mozafari, Z.; Noori, R.; Siadatmousavi, S.M.; Afzalimehr, H.; Azizpour, J. Satellite-based monitoring of eutrophication in the earth's largest transboundary lake. *GeoHealth* **2023**, *7*, e2022GH000770. [CrossRef] [PubMed]
74. Kupssinskü, L.S.; Guimarães, T.T.; de Souza, E.M.; Zanotta, D.C.; Veronez, M.R.; Gonzaga, L.; Mauad, F.F. A Method for Chlorophyll-a and Suspended Solids Prediction through Remote Sensing and Machine Learning. *Sensors* **2020**, *20*, 2125. [CrossRef] [PubMed]

75. Yang, M.; Ishizaka, J.; Goes, J.; Gomes, H.; Maúre, E.; Hayashi, M.; Katano, T.; Fujii, N.; Saitoh, K.; Mine, T.; et al. Improved MODIS-Aqua Chlorophyll-a Retrievals in the Turbid Semi-Enclosed Ariake Bay, Japan. *Remote Sens.* **2018**, *10*, 1335. [CrossRef]
76. Yang, Y.; Zhang, X.; Gao, W.; Zhang, Y.; Hou, X. Improving lake chlorophyll-a interpreting accuracy by combining spectral and texture features of remote sensing. *Environ. Sci. Pollut. Res.* **2023**, *30*, 83628–83642. [CrossRef]
77. Lu, L.; Gong, Z.; Liang, Y.; Liang, S. Retrieval of Chlorophyll-a Concentrations of Class II Water Bodies of Inland Lakes and Reservoirs Based on ZY1-02D Satellite Hyperspectral Data. *Remote Sens.* **2022**, *14*, 1842. [CrossRef]
78. Avdan, Z.Y.; Kaplan, G.; Goncu, S.; Avdan, U. Monitoring the water quality of small water bodies using high-resolution remote sensing data. *ISPRS Int. J. Geo-Inf.* **2019**, *8*, 553. [CrossRef]
79. Assar, W. Retrieval of Chlorophyll-a and vegetation indices Using Sentinel-2 MSI Imagery in El-Burullus Lake, Egypt. *SVU-Int. J. Eng. Sci. Appl.* **2024**, *5*, 79–88. [CrossRef]
80. Shi, J.; Shen, Q.; Yao, Y.; Li, J.; Chen, F.; Wang, R.; Xu, W.; Gao, Z.; Wang, L. Estimation of Chlorophyll-a Concentrations in Small Water Bodies: Comparison of Fused Gaofen-6 and Sentinel-2 Sensors. *Remote Sens.* **2022**, *14*, 229. [CrossRef]
81. Shi, X.; Gu, L.; Jiang, T.; Zheng, X.; Dong, W.; Tao, Z. Retrieval of Chlorophyll-a Concentrations Using Sentinel-2 MSI Imagery in Lake Chagan Based on Assessments with Machine Learning Models. *Remote Sens.* **2022**, *14*, 4924. [CrossRef]
82. Wang, D.; Tang, B.; Fu, Z.-G.; Huang, L.; Li, M.; Chen, G.; Pan, X. Estimation of Chlorophyll-A Concentration with Remotely Sensed Data for the Nine Plateau Lakes in Yunnan Province. *Remote Sens.* **2022**, *14*, 4950. [CrossRef]
83. Wu, C.; Fu, X.; Li, H.; Hu, H.; Li, X.; Zhang, L. A Method Based on Improved Ant Colony Algorithm Feature Selection Combined With GA-SVR Model for Predicting Chlorophyll-a Concentration in Ulansuhai Lake. *IEEE Access* **2023**, *11*, 93180–93192. [CrossRef]
84. Barraza-Moraga, F.; Alcayaga, H.; Pizarro, A.; Félez-Bernal, J.; Urrutia, R. Estimation of Chlorophyll-a Concentrations in Lanalhue Lake Using Sentinel-2 MSI Satellite Images. *Remote Sens.* **2022**, *14*, 5647. [CrossRef]
85. Pershin, S.; Katsnelson, B.; Grishin, M.; Lednev, V.; Zavozin, V.A.; Ostrovsky, I. Laser Remote Sensing of Lake Kinneret by Compact Fluorescence LiDAR. *Sensors* **2022**, *22*, 7307. [CrossRef]
86. Warren, M.A.; Simis, S.G.; Martinez-Vicente, V.; Poser, K.; Bresciani, M.; Alikas, K.; Spyrakos, E.; Giardino, C.; Ansper, A. Assessment of atmospheric correction algorithms for the Sentinel-2A MultiSpectral Imager over coastal and inland waters. *Remote Sens. Environ.* **2019**, *225*, 267–289. [CrossRef]
87. Ma, Y.; Song, K.; Wen, Z.; Liu, G.; Shang, Y.; Lyu, L.; Du, J.; Yang, Q.; Li, S.; Tao, H.; et al. Remote sensing of turbidity for lakes in northeast China using Sentinel-2 images with machine learning algorithms. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* **2021**, *14*, 9132–9146. [CrossRef]
88. Sakuno, Y.; Yajima, H.; Yoshioka, Y.; Sugahara, S.; Abd Elbasit, M.A.; Adam, E.; Chirima, J.G. Evaluation of unified algorithms for remote sensing of chlorophyll-a and turbidity in lake Shinji and lake Nakaumi of Japan and the Vaal Dam reservoir of South Africa under eutrophic and ultra-turbid conditions. *Water* **2018**, *10*, 618. [CrossRef]
89. Villota-González, F.H.; Sulbarán-Rangel, B.; Zurita-Martínez, F.; Gurubel-Tun, K.J.; Zúñiga-Grajeda, V. Assessment of Machine Learning Models for Remote Sensing of Water Quality in Lakes Cajititlán and Zapotlán, Jalisco—Mexico. *Remote Sens.* **2023**, *15*, 5505. [CrossRef]
90. Pamula, A.; Gholizadeh, H.; Krzmarzick, M.; Mausbach, W.; Lampert, D. A remote sensing tool for near real-time monitoring of harmful algal blooms and turbidity in reservoirs. *JAWRA J. Am. Water Resour. Assoc.* **2023**, *59*, 929–949. [CrossRef]
91. Cui, M.; Sun, Y.; Huang, C.; Li, M. Water turbidity retrieval based on UAV hyperspectral remote sensing. *Water* **2022**, *14*, 128. [CrossRef]
92. Rodríguez-López, L.; Duran-Llacer, I.; González-Rodríguez, L.; Cárdenas, R.; Urrutia, R. Retrieving Water Turbidity in Araucanian Lakes (South-Central Chile) Based on Multispectral Landsat Imagery. *Remote Sens.* **2021**, *13*, 3133. [CrossRef]
93. Ramesh, J.V.N.; Patibandla, P.R.; Shanbhog, M.; Ambala, S.; Ashraf, M.; Kiran, A. Ensemble deep learning approach for turbidity prediction of Dooskal Lake using remote sensing data. *Remote Sens. Earth Syst. Sci.* **2023**, *6*, 146–155. [CrossRef]
94. Liu, G.; Li, S.; Song, K.; Wang, X.; Wen, Z.; Kutser, T.; Jacinthe, P.-A.; Shang, Y.; Lyu, L.; Fang, C.; et al. Remote sensing of CDOM and DOC in alpine lakes across the Qinghai-Tibet Plateau using Sentinel-2A imagery data. *J. Environ. Manag.* **2021**, *286*, 112231. [CrossRef]
95. Zhang, H.; Yao, B.; Wang, S.; Wang, G. Remote sensing estimation of the concentration and sources of coloured dissolved organic matter based on MODIS: A case study of Erhai lake. *Ecol. Indic.* **2021**, *131*, 108180. [CrossRef]
96. Hulley, G.C.; Hook, S.J. Generating consistent land surface temperature and emissivity products between ASTER and MODIS data for earth science research. *IEEE Trans. Geosci. Remote Sens.* **2011**, *49*, 1304–1315. [CrossRef]
97. Dyba, K.; Ermida, S.; Ptak, M.; Piekarczyk, J.; Sojka, M. Evaluation of methods for estimating lake surface water temperature using Landsat 8. *Remote Sens.* **2022**, *14*, 3839. [CrossRef]
98. Yuan, K.; Zhuang, X.; Schaefer, G.; Feng, J.; Guan, L.; Fang, H. Deep-Learning-Based Multispectral Satellite Image Segmentation for Water Body Detection. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* **2021**, *14*, 7422–7434. [CrossRef]
99. Wu, Y.; Han, P.; Zheng, Z. Instant water body variation detection via analysis on remote sensing imagery. *J. Real-Time Image Process.* **2021**, *18*, 1577–1590. [CrossRef]
100. Lin, Y.; Li, X.; Zhang, T.; Chao, N.; Yu, J.; Cai, J.; Sneeuw, N. Water volume variations estimation and analysis using multisource satellite data: A case study of Lake Victoria. *Remote Sens.* **2020**, *12*, 3052. [CrossRef]
101. Bijeesh, T.; Narasimhamurthy, K. Surface water detection and delineation using remote sensing images: A review of methods and algorithms. *Sustain. Water Resour. Manag.* **2020**, *6*, 68. [CrossRef]

102. Zhang, Y.; Gu, Z.; Thé, J.V.G.; Yang, S.X.; Gharabaghi, B. The discharge forecasting of multiple monitoring station for humber river by hybrid LSTM models. *Water* **2022**, *14*, 1794. [CrossRef]
103. Chipman, J.W. A multisensor approach to satellite monitoring of trends in lake area, water level, and volume. *Remote Sens.* **2019**, *11*, 158. [CrossRef]
104. Song, Q.; Zhao, R.; Fu, H.; Zhu, J.; Li, Y. Water Area Extraction and Water Level Prediction of Dongting Lake Based on Sentinel-1 Dual-Polarization Data Decomposition. *Remote Sens.* **2023**, *15*, 4655. [CrossRef]
105. Nan, Y.; Jianhui, L.; Wenbo, M.; Wangjun, L.; Di, W.; Wanchao, G.; Changhao, S. Water depth retrieval models of East Dongting Lake, China, using GF-1 multi-spectral remote sensing images. *Glob. Ecol. Conserv.* **2020**, *22*, e01004. [CrossRef]
106. Naji, T.A.H.; Abdul-Hameed, S.A.Q.; Abdul-Hameed, D.A.Q.; Awad, T.M.; Fakhruldeen, S.S.R. The use of remote sensing technology in defining the water depth in the lakes and water bodies: Western Iraq as a case study. *J. Glob. Innov. Agric. Sci.* **2023**, *11*, 191–197. [CrossRef]
107. Song, W.; Yinglan, A.; Wang, Y.; Fang, Q.; Tang, R. Study on remote sensing inversion and temporal-spatial variation of Hulun lake water quality based on machine learning. *J. Contam. Hydrol.* **2024**, *260*, 104282. [CrossRef]
108. Wang, W.; Yang, P.; Xia, J.; Zhang, S.; Luo, X.; Hu, S.; Zhan, C. Characterizing water body changes in Poyang lake using multi-source remote sensing data. *Environ. Dev.* **2023**, *48*, 100909. [CrossRef]
109. Yang, H.; Guo, H.; Dai, W.; Nie, B.; Qiao, B.; Zhu, L. Bathymetric mapping and estimation of water storage in a shallow lake using a remote sensing inversion method based on machine learning. *Int. J. Digit. Earth* **2022**, *15*, 789–812. [CrossRef]
110. Jakovljevic, G.; Álvarez-Taboada, F.; Govedarica, M. Long-term monitoring of inland water quality parameters using Landsat time-series and back-propagated ANN: Assessment and usability in a real-case scenario. *Remote Sens.* **2023**, *16*, 68. [CrossRef]
111. Zhang, H.; Xue, B.; Wang, G.; Zhang, X.; Zhang, Q. Deep Learning-Based Water Quality Retrieval in an Impounded Lake Using Landsat 8 Imagery: An Application in Dongping Lake. *Remote Sens.* **2022**, *14*, 4505. [CrossRef]
112. Chen, J.; Zhao, Q. An approach for estimating MODIS reflectance at top-of-atmosphere at 667 and 678 nm from reflectance at 645 nm in turbid waters. *Environ. Earth Sci.* **2014**, *71*, 1105–1114. [CrossRef]
113. Yang, W.; Matsushita, B.; Chen, J.; Yoshimura, K.; Fukushima, T. Retrieval of inherent optical properties for turbid inland waters from remote-sensing reflectance. *IEEE Trans. Geosci. Remote Sens.* **2013**, *51*, 3761–3773. [CrossRef]
114. Sayers, M.; Bosse, K.; Shuchman, R.; Ruberg, S.; Fahnenstiel, G.; Leshkevich, G.; Stuart, D.; Johengen, T.; Burtner, A.; Palladino, D. Spatial and temporal variability of inherent and apparent optical properties in western Lake Erie: Implications for water quality remote sensing. *J. Great Lakes Res.* **2019**, *45*, 490–507. [CrossRef]
115. Matsui, K.; Shirai, H.; Kageyama, Y.; Yokoyama, H. Improving the resolution of UAV-based remote sensing data of water quality of Lake Hachiroko, Japan by neural networks. *Ecol. Inform.* **2021**, *62*, 101276. [CrossRef]
116. Cillero Castro, C.; Domínguez Gómez, J.A.; Delgado Martín, J.; Hinojo Sánchez, B.A.; Cereijo Arango, J.L.; Cheda Tuya, F.A.; Díaz-Varela, R. An UAV and satellite multispectral data approach to monitor water quality in small reservoirs. *Remote Sens.* **2020**, *12*, 1514. [CrossRef]
117. Rahul, T.S.; Brema, J.; Wessley, G.J.J. Evaluation of surface water quality of Ukkadam lake in Coimbatore using UAV and Sentinel-2 multispectral data. *Int. J. Environ. Sci. Technol.* **2023**, *20*, 3205–3220. [CrossRef]
118. Lo, Y.; Fu, L.; Lu, T.; Huang, H.; Kong, L.; Xu, Y.; Zhang, C. Medium-sized lake water quality parameters retrieval using multispectral UAV image and machine learning algorithms: A case study of the Yuandang Lake, China. *Drones* **2023**, *7*, 244. [CrossRef]
119. Guimarães, T.T.; Veronez, M.R.; Koste, E.C.; Souza, E.M.; Brum, D.; Gonzaga, L., Jr.; Mauad, F.F. Evaluation of regression analysis and neural networks to predict total suspended solids in water bodies from unmanned aerial vehicle images. *Sustainability* **2019**, *11*, 2580. [CrossRef]
120. Yang, H.; Kong, J.; Hu, H.; Du, Y.; Gao, M.; Chen, F. A review of remote sensing for water quality retrieval: Progress and challenges. *Remote Sens.* **2022**, *14*, 1770. [CrossRef]
121. Strong, A.E. Remote sensing of algal blooms by aircraft and satellite in Lake Erie and Utah Lake. *Remote Sens. Environ.* **1974**, *3*, 99–107. [CrossRef]
122. Vogt, M.C.; Vogt, M.E. Near-remote sensing of water turbidity using small unmanned aircraft systems. *Environ. Pract.* **2016**, *18*, 18–31. [CrossRef]
123. Guo, H.; Tian, S.; Huang, J.J.; Zhu, X.; Wang, B.; Zhang, Z. Performance of deep learning in mapping water quality of Lake Simcoe with long-term Landsat archive. *ISPRS J. Photogramm. Remote Sens.* **2022**, *183*, 451–469. [CrossRef]
124. Zhu, W.; Huang, L.; Sun, N.; Chen, J.; Pang, S. Landsat 8-observed water quality and its coupled environmental factors for urban scenery lakes: A case study of West Lake. *Water Environ. Res.* **2020**, *92*, 255–265. [CrossRef] [PubMed]
125. Soomets, T.; Uudeberg, K.; Jakovels, D.; Brauns, A.; Zagars, M.; Kutser, T. Validation and comparison of water quality products in Baltic lakes using Sentinel-2 MSI and Sentinel-3 OLCI data. *Sensors* **2020**, *20*, 742. [CrossRef]
126. Ambrose-Igho, G.; Seyoum, W.M.; Perry, W.L.; O'Reilly, C.M. Spatiotemporal analysis of water quality indicators in small lakes using Sentinel-2 satellite data: Lake Bloomington and Evergreen Lake, Central Illinois, USA. *Environ. Process.* **2021**, *8*, 637–660. [CrossRef]
127. Wang, S.; Li, J.; Zhang, B.; Lee, Z.; Spyrakos, E.; Feng, L.; Liu, C.; Zhao, H.; Wu, Y.; Zhu, L.; et al. Changes of water clarity in large lakes and reservoirs across China observed from long-term MODIS. *Remote Sens. Environ.* **2020**, *247*, 111949. [CrossRef]

128. Binding, C.E.; Greenberg, T.A.; Bukata, R.P. An analysis of MODIS-derived algal and mineral turbidity in Lake Erie. *J. Great Lakes Res.* **2012**, *38*, 107–116. [CrossRef]
129. Deutsch, E.; Alameddine, I.; El-Fadel, M. Monitoring water quality in a hypereutrophic reservoir using Landsat ETM+ and OLI sensors: How transferable are the water quality algorithms? *Environ. Monit. Assess.* **2018**, *190*, 141. [CrossRef]
130. Chen, J.; Zhu, W. Consistency evaluation of Landsat-7 and Landsat-8 for improved monitoring of colored dissolved organic matter in complex water. *Geocarto Int.* **2020**, *37*, 91–102. [CrossRef]
131. Masocha, M.; Dube, T.; Nhiwatiwa, T.; Choruma, D. Testing utility of Landsat 8 for remote assessment of water quality in two subtropical African reservoirs with contrasting trophic states. *Geocarto Int.* **2018**, *33*, 667–680. [CrossRef]
132. Markogianni, V.; Kalivas, D.; Petropoulos, G.P.; Dimitriou, E. An appraisal of the potential of Landsat 8 in estimating chlorophyll-a, ammonium concentrations and other water quality indicators. *Remote Sens.* **2018**, *10*, 1018. [CrossRef]
133. Sobrino, J.A.; García-Monteiro, S.; Julien, Y. An analysis of the Lake Surface Water Temperature evolution of the world's largest lakes during the years 2003–2020 using MODIS data. *Recent Adv. Remote Sens.* **2024**, *1*, 1–9. [CrossRef]
134. Xie, C.; Zhang, X.; Zhuang, L.; Zhu, R.; Guo, J. Analysis of surface temperature variation of lakes in China using MODIS land surface temperature data. *Sci. Rep.* **2022**, *12*, 2415. [CrossRef] [PubMed]
135. Zhao, G.; Gao, H.; Cai, X. Estimating lake temperature profile and evaporation losses by leveraging MODIS LST data. *Remote Sens. Environ.* **2020**, *251*, 112104. [CrossRef]
136. Tavares, M.H.; Cardoso, M.A.; Motta Marques, D.; Bonnet, M.P.; Fragoso, C.R., Jr. High Spring Warming Rates in an Extensive Subtropical Shallow Lakes System Detected Using MODIS Imagery. *ACS EST Water* **2024**, *4*, 1518–1530. [CrossRef]
137. Toming, K.; Kutser, T.; Laas, A.; Sepp, M.; Paavel, B.; Nõges, T. First experiences in mapping lake water quality parameters with Sentinel-2 MSI imagery. *Remote Sens.* **2016**, *8*, 640. [CrossRef]
138. Al-Kharusi, E.S.; Tenenbaum, D.E.; Abdi, A.M.; Kutser, T.; Karlsson, J.; Bergström, A.K.; Berggren, M. Large-scale retrieval of coloured dissolved organic matter in northern lakes using Sentinel-2 data. *Remote Sens.* **2020**, *12*, 157. [CrossRef]
139. Pahlevan, N.; Chittimalli, S.K.; Balasubramanian, S.V.; Vellucci, V. Sentinel-2/Landsat-8 product consistency and implications for monitoring aquatic systems. *Remote Sens. Environ.* **2019**, *220*, 19–29. [CrossRef]
140. Bonansea, M.; Ledesma, M.; Bazán, R.; Ferral, A.; German, A.; O'Mill, P.; Rodriguez, C.; Pinotti, L. Evaluating the feasibility of using Sentinel-2 imagery for water clarity assessment in a reservoir. *J. S. Am. Earth Sci.* **2019**, *95*, 102265. [CrossRef]
141. Sent, G.; Biguino, B.; Favareto, L.; Cruz, J.; Sá, C.; Dogliotti, A.I.; Palma, C.; Brotas, V.; Brito, A.C. Deriving water quality parameters using sentinel-2 imagery: A case study in the Sado Estuary, Portugal. *Remote Sens.* **2021**, *13*, 1043. [CrossRef]
142. Guo, H.; Huang, J.; Chen, B.; Guo, X.; Singh, V. A machine learning-based strategy for estimating non-optically active water quality parameters using Sentinel-2 imagery. *Int. J. Remote Sens.* **2020**, *42*, 1841–1866. [CrossRef]
143. Giardino, C.; Brando, V.; Dekker, A.; Strömbeck, N.; Candiani, G. Assessment of water quality in Lake Garda (Italy) using Hyperion. *Remote Sens. Environ.* **2007**, *109*, 183–195. [CrossRef]
144. Wang, S.; Yan, F.; Zhou, Y.; Zhu, L.; Wang, L.; Jiao, Y. Water quality monitoring using hyperspectral remote sensing data in Taihu Lake China. In Proceedings of the 2005 IEEE International Geoscience and Remote Sensing Symposium, IGARSS '05, Seoul, Republic of Korea, 29 July 2005; Volume 7, pp. 4553–4556.
145. Brando, V.; Dekker, A. Satellite hyperspectral remote sensing for estimating estuarine and coastal water quality. *IEEE Trans. Geosci. Remote Sens.* **2003**, *41*, 1378–1387. [CrossRef]
146. Kaplan, G.; Avdan, Z.Y.; Goncu, S.; Avdan, U. Evaluation of RapidEye-3 Satellite Data for Assessing Water Turbidity of Lake Borabey. *Proceedings* **2019**, *48*, 14. [CrossRef]
147. Mishra, S.; Stumpf, R.; Meredith, A. Evaluation of RapidEye data for mapping algal blooms in inland waters. *Int. J. Remote Sens.* **2018**, *40*, 2811–2829. [CrossRef]
148. Su, P.W.; Lo, S.L. Using Landsat 8 imagery for remote monitoring of total phosphorus as a water quality parameter of irrigation ponds in Taiwan. *Environ. Sci. Pollut. Res.* **2021**, *28*, 66687–66694. [CrossRef]
149. Wu, M.; Zhang, W.; Wang, X.; Luo, D. Application of MODIS satellite data in monitoring water quality parameters of Chaohu Lake in China. *Environ. Monit. Assess.* **2009**, *148*, 255–264. [CrossRef]
150. Lv, H.; Huang, J.Z.; Jiang, N. Quantitative retrieval of water quality by remote sensing in Taihu Lake based on MODIS data. *J. Geo-Inf. Sci.* **2009**, *11*, 104–110.
151. Xiaoyu, Z.; Dingtian, Y.; Xiaofeng, Z.; Difeng, W.; Shujing, L.; Delu, P. Pre-Operational Monitoring of Large Inland Lake Water Quality by MODIS Imagery. In Proceedings of the SPIE Remote Sensing of the Ocean, Sea Ice, and Large Water Regions, Bruges, Belgium, 19–22 September 2005; Volume 5977, pp. 148–157.
152. Duan, H.; Tao, M.; Loiselle, S.; Zhao, W.; Cao, Z.; Ma, R.; Tang, X. MODIS observations of cyanobacterial risks in a eutrophic lake: Implications for long-term safety evaluation in drinking-water source. *Water Res.* **2017**, *122*, 455–470. [CrossRef]
153. Balasubramanian, S.; Pahlevan, N.; Smith, B.; Binding, C.; Schalles, J.; Loisel, H.; Gurlin, D.; Greb, S.; Alikas, K.; Randla, M.; et al. Robust algorithm for estimating total suspended solids (TSS) in inland and nearshore coastal waters. *Remote Sens. Environ.* **2020**, *246*, 111768. [CrossRef]
154. Liu, X.; Zhang, Z.; Jiang, T.; Li, X.; Li, Y. Evaluation of the Effectiveness of Multiple Machine Learning Methods in Remote Sensing Quantitative Retrieval of Suspended Matter Concentrations: A Case Study of Nansi Lake in North China. *J. Spectrosc.* **2021**, *2021*, 5957376. [CrossRef]

155. Ren, J.; Zhou, H.; Tao, Z.; Ge, L.; Song, K.; Xu, S.; Li, Y.; Zhang, L.; Zhang, X.; Li, S. Long-term monitoring chlorophyll-a concentration using HJ-1 A/B imagery and machine learning algorithms in typical lakes, a cold semi-arid region. *Opt. Express* **2024**, *32*, 16371–16397. [CrossRef]
156. Kontopoulou, E.; Kolokoussis, P.; Karantzalos, K. Water quality estimation in Greek lakes from Landsat 8 multispectral satellite data. *Eur. Water* **2017**, *58*, 191–196.
157. Al-Shaibah, B.; Liu, X.; Zhang, J.; Tong, Z.; Zhang, M.; El-Zeiny, A.; Faichia, C.; Hussain, M.; Tayyab, M. Modeling water quality parameters using landsat multispectral images: A case study of Erlong lake, northeast China. *Remote Sens.* **2021**, *13*, 1603. [CrossRef]
158. Yang, Z.; Gong, C.; Ji, T.; Hu, Y.; Li, L. Water Quality Retrieval from ZY1-02D Hyperspectral Imagery in Urban Water Bodies and Comparison with Sentinel-2. *Remote Sens.* **2022**, *14*, 5029. [CrossRef]
159. Niroumand-Jadidi, M.; Bovolo, F.; Bruzzone, L. Water quality retrieval from PRISMA hyperspectral images: First experience in a turbid lake and comparison with Sentinel-2. *Remote Sens.* **2020**, *12*, 3984. [CrossRef]
160. Zhang, D.; Zhang, L.; Sun, X.; Gao, Y.; Lan, Z.; Wang, Y.; Zhai, H.; Li, J.; Wang, W.; Chen, M.; et al. A new method for calculating water quality parameters by integrating space–ground hyperspectral data and spectral-in situ assay data. *Remote Sens.* **2022**, *14*, 3652. [CrossRef]
161. Yang, J.; Zhou, A.; Han, L.; Li, Y.; Xie, Y. Monitoring urban black-odorous water by using hyperspectral data and machine learning. *Environ. Pollut.* **2021**, *269*, 116166.
162. Singh, A.; Jakubowski, A.; Chidister, I.; Townsend, P. A MODIS approach to predicting stream water quality in Wisconsin. *Remote Sens. Environ.* **2013**, *128*, 74–86. [CrossRef]
163. Rodríguez-López, L.; Duran-Llacer, I.; González-Rodríguez, L.; Abarca-del-Río, R.; Cárdenas, R.; Parra, O.; Martínez-Retureta, R.; Urrutia, R. Spectral analysis using LANDSAT images to monitor the chlorophyll-a concentration in Lake Laja in Chile. *Ecol. Inform.* **2020**, *60*, 101183. [CrossRef]
164. Govil, H.; Guha, S.; Diwan, P.; Gill, N.; Dey, A. Analyzing Linear Relationships of LST with NDVI and MNDISI Using Various Resolution Levels of Landsat 8 OLI and TIRS Data. In *Data Management, Analytics and Innovation*; Springer: Berlin/Heidelberg, Germany, 2019.
165. Wang, X.; Song, K.; Wen, Z.; Liu, G.; Shang, Y.; Fang, C.; Lyu, L.; Wang, Q. Quantifying Turbidity Variation for Lakes in Daqing of Northeast China Using Landsat Images From 1984 to 2018. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* **2021**, *14*, 8884–8897. [CrossRef]
166. McFeeters, S. The use of the Normalized Difference Water Index (NDWI) in the delineation of open water features. *Int. J. Remote Sens.* **1996**, *17*, 1425–1432. [CrossRef]
167. Helali, J.; Asaadi, S.; Jafarie, T.; Habibi, M.; Salimi, S.; Momenpour, S.; Shahmoradi, S.; Hosseini, S.; Hessari, B.; Saeidi, V. Drought monitoring and its effects on vegetation and water extent changes using remote sensing data in Urmia Lake watershed, Iran. *J. Water Clim. Change* **2022**, *13*, 2107–2128. [CrossRef]
168. Elhag, M.; Gitas, I.; Othman, A.; Bahrawi, J.; Gikas, P. Assessment of water quality parameters using temporal remote sensing spectral reflectance in arid environments, Saudi Arabia. *Water* **2019**, *11*, 556. [CrossRef]
169. Özelkan, E. Water body detection analysis using NDWI indices derived from Landsat-8 OLI. *Pol. J. Environ. Stud.* **2020**, *29*, 1759–1769. [CrossRef]
170. Mishra, S.; Mishra, D. Normalized difference chlorophyll index: A novel model for remote estimation of chlorophyll-a concentration in turbid productive waters. *Remote Sens. Environ.* **2012**, *117*, 394–406. [CrossRef]
171. Zhang, F.; Zhang, B.; Li, J.; Shen, Q.; Wu, Y.; Wang, G.; Zou, L.; Wang, S. Validation of a synthetic chlorophyll index for remote estimates of chlorophyll-a in a turbid hypereutrophic lake. *Int. J. Remote Sens.* **2014**, *35*, 289–305. [CrossRef]
172. Watanabe, F.; Alcantara, E.; Rodrigues, T.; Rotta, L.; Bernardo, N.; Imai, N. Remote sensing of the chlorophyll-a based on OLI/Landsat-8 and MSI/Sentinel-2A (Barra Bonita reservoir, Brazil). *An. Acad. Bras. Ciências* **2017**, *90* (Suppl. S1), 1987–2000. [CrossRef]
173. Malahlela, O.E. Spatio-temporal assessment of inland surface water quality using remote sensing data in the wake of changing climate. *IOP Conf. Ser. Earth Environ. Sci.* **2019**, *227*, 062012. [CrossRef]
174. Bishop, C.M.; Nasrabadi, N.M. *Pattern Recognition and Machine Learning*; Springer: New York, NY, USA, 2006; Volume 4, p. 738.
175. Goodfellow, I.; Bengio, Y.; Courville, A. *Deep Learning*; MIT Press: Cambridge, MA, USA, 2016.
176. Hajigholizadeh, M.; Moncada, A.; Kent, S.; Melesse, A.M. Land–lake linkage and remote sensing application in water quality monitoring in Lake Okeechobee, Florida, USA. *Land* **2021**, *10*, 147. [CrossRef]
177. Wang, N.; Wang, Z.; Huang, P.; Zhai, Y.; Yang, X.; Su, J. Inversion Method for Chlorophyll-a Concentration in High-Salinity Water Based on Hyperspectral Remote Sensing Data. *Sensors* **2024**, *24*, 4181. [CrossRef]
178. Zhao, Y.; Yu, T.; Hu, B.; Zhang, Z.; Liu, Y.; Liu, X.; Liu, H.; Liu, J.; Wang, X.; Song, S. Retrieval of water quality parameters based on near-surface remote sensing and machine learning algorithm. *Remote Sens.* **2022**, *14*, 5305. [CrossRef]
179. Qiao, Z.; Sun, S.; Jiang, Q.O.; Xiao, L.; Wang, Y.; Yan, H. Retrieval of total phosphorus concentration in the surface water of Miyun reservoir based on remote sensing data and machine learning algorithms. *Remote Sens.* **2021**, *13*, 4662. [CrossRef]
180. Qun'ou, J.; Lidan, X.; Siyang, S.; Meilin, W.; Huijie, X. Retrieval model for total nitrogen concentration based on UAV hyperspectral remote sensing data and machine learning algorithms—A case study in the Miyun Reservoir, China. *Ecol. Indic.* **2021**, *124*, 107356. [CrossRef]

181. Mu, M.; Li, Y.; Bi, S.; Lyu, H.; Xu, J.; Lei, S.; Miao, S.; Zeng, S.; Zheng, Z.; Du, C. Prediction of algal bloom occurrence based on the naive Bayesian model considering satellite image pixel differences. *Ecol. Indic.* **2021**, *124*, 107416. [CrossRef]
182. Peng, C.; Xie, Z.; Jin, X. Using Ensemble Learning for Remote Sensing Inversion of Water Quality Parameters in Poyang Lake. *Sustainability* **2024**, *16*, 3355. [CrossRef]
183. Leggesse, E.S.; Zimale, F.A.; Sultan, D.; Enku, T.; Srinivasan, R.; Tilahun, S.A. Predicting optical water quality indicators from remote sensing using machine learning algorithms in tropical highlands of Ethiopia. *Hydrology* **2023**, *10*, 110. [CrossRef]
184. Zhang, Y.; Wu, L.; Ren, H.; Deng, L.; Zhang, P. Retrieval of water quality parameters from hyperspectral images using hybrid Bayesian probabilistic neural network. *Remote Sens.* **2020**, *12*, 1567. [CrossRef]
185. Arias-Rodriguez, L.F.; Duan, Z.; de Jesús Díaz-Torres, J.; Basilio Hazas, M.; Huang, J.; Kumar, B.U.; Tuo, Y.; Disse, M. Integration of remote sensing and Mexican water quality monitoring system using an extreme learning machine. *Sensors* **2021**, *21*, 4118. [CrossRef]
186. Meng, H.; Zhang, J.; Zheng, Z.; Song, Y.; Lai, Y. Classification of inland lake water quality levels based on Sentinel-2 images using convolutional neural networks and spatiotemporal variation and driving factors of algal bloom. *Ecol. Inform.* **2024**, *80*, 102549. [CrossRef]
187. Giménez, J.G.; González, M.; Martínez-España, R.; Cecilia, J.M.; López-Espín, J.J. Enhancing shallow water quality monitoring efficiency with deep learning and remote sensing: A case study in Mar Menor. *J. Ambient. Intell. Smart Environ.* **2024**. [CrossRef]
188. Liu, M.; Liu, J.; Hu, H. A Novel Deep Learning Network Model for Extracting Lake Water Bodies from Remote Sensing Images. *Appl. Sci.* **2024**, *14*, 1344. [CrossRef]
189. Rodríguez-López, L.; Bustos Usta, D.; Duran-Llacer, I.; Bravo Alvarez, L.; Yépez, S.; Bourrel, L.; Frappart, F.; Urrutia, R. Estimation of Water Quality Parameters through a Combination of Deep Learning and Remote Sensing Techniques in a Lake in Southern Chile. *Remote Sens.* **2023**, *15*, 4157. [CrossRef]
190. Zhang, Y.; Zhou, Z.; Van Griensven Thé, J.; Yang, S.X.; Gharabaghi, B. Flood Forecasting Using Hybrid LSTM and GRU Models with Lag Time Preprocessing. *Water* **2023**, *15*, 3982. [CrossRef]
191. Tang, Y.; Feng, Y.; Fung, S.; Xomchuk, V.R.; Jiang, M.; Moore, T.; Beckler, J. Spatiotemporal deep-learning-based algal bloom prediction for Lake Okeechobee using multisource data fusion. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* **2022**, *15*, 8318–8331. [CrossRef]
192. Pan, D.; Zhang, Y.; Deng, Y.; Van Griensven Thé, J.; Yang, S.X.; Gharabaghi, B. Dissolved Oxygen Forecasting for Lake Erie's Central Basin Using Hybrid Long Short-Term Memory and Gated Recurrent Unit Networks. *Water* **2024**, *16*, 707. [CrossRef]
193. Pan, B.; Yu, H.; Cheng, H.; Du, S.; Cai, S.; Zhao, M.; Du, J.; Xie, F. A CNN–LSTM Machine-Learning Method for Estimating Particulate Organic Carbon from Remote Sensing in Lakes. *Sustainability* **2023**, *15*, 13043. [CrossRef]
194. Zhang, Y.; Zhou, Z.; Deng, Y.; Pan, D.; Van Griensven Thé, J.; Yang, S.X.; Gharabaghi, B. Daily Streamflow Forecasting Using Networks of Real-Time Monitoring Stations and Hybrid Machine Learning Methods. *Water* **2024**, *16*, 1284. [CrossRef]
195. Ramaraj, M.; Sivakumar, R. Integration of band regression empirical water quality (BREWQ) model with deep learning algorithm in spatiotemporal modeling and prediction of surface water quality parameters. *Model. Earth Syst. Environ.* **2023**, *9*, 3279–3304. [CrossRef]
196. Hao, Z.; Li, W.; Wu, J.; Zhang, S.; Hu, S. A novel deep learning model for mining nonlinear dynamics in lake surface water temperature prediction. *Remote Sens.* **2023**, *15*, 900. [CrossRef]
197. Huynh, N.H.; Böer, G.; Schramm, H. Self-attention and generative adversarial networks for algae monitoring. *Eur. J. Remote Sens.* **2022**, *55*, 10–22. [CrossRef]
198. Wang, L.; Li, W.; Wang, X.; Xu, J. Remote sensing image analysis and prediction based on improved Pix2Pix model for water environment protection of smart cities. *PeerJ Comput. Sci.* **2023**, *9*, e1292. [CrossRef] [PubMed]
199. Zhang, Y.; Pan, D.; Van Griensven, J.; Yang, S.X.; Gharabaghi, B. Intelligent flood forecasting and warning: A survey. *Intell. Robot.* **2023**, *3*, 190–212. [CrossRef]
200. Maxwell, A.; Sharma, M.; Donaldson, K. Explainable Boosting Machines for Slope Failure Spatial Predictive Modeling. *Remote Sens.* **2021**, *13*, 4991. [CrossRef]
201. Srivastava, N.; Hinton, G.; Krizhevsky, A.; Sutskever, I.; Salakhutdinov, R. Dropout: A simple way to prevent neural networks from overfitting. *J. Mach. Learn. Res.* **2014**, *15*, 1929–1958. [CrossRef]
202. Alqahtani, A.; Shah, M.; Aldrees, A.; Javed, M. Comparative assessment of individual and ensemble machine learning models for efficient analysis of river water quality. *Sustainability* **2022**, *14*, 1183. [CrossRef]
203. Devi, P.D.; Mamatha, G. Machine learning approach to predict the turbidity of Saki Lake, Telangana, India, using remote sensing data. *Meas. Sens.* **2024**, *33*, 101139. [CrossRef]
204. Xu, M. Assessment Method of Lake's Water Quality Based on Remote-Sensed Image and Support Vector Machine. In Proceedings of the IEEE Conference on Industrial Electronics and Applications, Harbin, China, 15–17 May 2007; pp. 2565–2570.
205. Hill, P.; Kumar, A.; Temimi, M.; Bull, D. HABNet: Machine learning, remote sensing-based detection of harmful algal blooms. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* **2020**, *13*, 3229–3239. [CrossRef]
206. Chang, N.B.; Bai, K.; Chen, C.F. Integrating multisensor satellite data merging and image reconstruction in support of machine learning for better water quality management. *J. Environ. Manag.* **2017**, *201*, 227–240. [CrossRef]
207. Sedighkia, M.; Datta, B.; Saeedipour, P.; Abdoli, A. Predicting water quality distribution of lakes through linking remote sensing-based monitoring and machine learning simulation. *Remote Sens.* **2023**, *15*, 3302. [CrossRef]

208. Li, Y.; Sun, Y.C.; Huang, J.; McBean, E. Retrieval of water quality parameters in Lake Ontario based on hyperspectral remote sensing data and intelligent algorithms. *Remote Sens.* **2020**, *14*, 1869. [CrossRef]
209. Lakshmi, G.; Bodapati, V.; Baji, S.; Panuganti, T. Water Quality Monitoring Using Remote-Sensing. In Proceedings of the 2023 International Conference on Computational Intelligence and Sustainable Engineering Solutions (CISES), Greater Noida, India, 28–30 April 2023; pp. 624–628.
210. Siddiqi, A.; Baber, S.; Weck, O. Valuing Radiometric Quality of Remote Sensing Data for Decisions. In Proceedings of the 2021 IEEE International Geoscience and Remote Sensing Sympsosium IGARSS, Brussels, Belgium, 11–16 July 2021; pp. 5724–5727.
211. Arias-Rodriguez, L.; Tüzün, U.; Duan, Z.; Huang, J.; Tuo, Y.; Disse, M. Global Water Quality of Inland Waters with Harmonized Landsat-8 and Sentinel-2 Using Cloud-Computed Machine Learning. *Remote Sens.* **2023**, *15*, 1390. [CrossRef]
212. Chen, J.; Fu, R.; Chen, S.; Li, D.; Jiang, H.; Wang, C.; Peng, Y.; Jia, K. Remote Sensing Big Data for Water Environment Monitoring: Current Status, Challenges, and Future Prospects. *Earth's Future* **2022**, *10*, e2021EF002289. [CrossRef]

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